Structural and Multidisciplinary Optimization (2022) 65: 245
https://doi.org/10.1007/s00158-022-03330-w

RESEARCH PAPER

Synergistic use of adjoint-based topology and shape optimization for the design of Bi-fluid heat exchangers

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Received: 14 February 2022 / Revised: 26 May 2022 / Accepted: 5 July 2022 / Published online: 18 August 2022
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Abstract

A method that combines topology (TopO) and shape (ShpO) optimization for the design of bi-fluid heat exchangers is presented. The porosity-based conjugate heat transfer (CHT) TopO creates flow paths for the two fluids (cooled and coolant) that should remain separated by solid material of adequate thickness. Heat transfer between the three media is taken into account by solving the energy equation for the two fluids and the heat conduction equation for the solid. A continuous porosity field is used and a Brinkman penalization term is introduced into the flow equations of each fluid to enforce an almost zero flow velocity over the solid area and the area where the other fluid flows. An initialization based on the low-cost numerical solution of a simplified Darcy-like TopO model, is proposed. Inaccuracies due to the weak treatment of flow conditions along the sought Fluid-Solid Interface (FSI) in TopO are overcome by, (a) a new interpolation scheme for the thermal conductivity at the cell faces, inspired by the CHT modeling with body-fitted grids, to be used during the TopO process and, (b) the successive implementation of TopO and ShpO. The latter requires that the TopO solution be post-processed to extract the FSI geometry and generate body-fitted grids. Both TopO and ShpO are supported by their corresponding continuous adjoint solvers in OpenFOAM, including the adjoint to the one-equation Spalart–Allmaras turbulence model as well as the adjoint to the Hamilton-Jacobi equation computing distances from the sought FSIs. The proposed method is demonstrated in the design of a compact bi-fluid heat exchanger.

Keywords Conjugate heat transfer · Multidisciplinary optimization · Topology optimization · Topology-to-shape transition · Shape optimization · Continuous adjoint method · Bi-fluid heat exchangers

1 Introduction

Heat exchangers are used in a huge variety of cooling and heating applications, Incopera et al. (2007). They operate with two different fluids, the cooled and the coolant, which should remain separated by solid material of adequate thickness. Their purpose is to exchange as much thermal energy as possible between the working fluids, with a low energy consumption to maintain the flow; these readily define the objectives in any relevant optimization run. The performance of heat exchangers highly depends on the shape of the flow domain. The number of flow channels for each fluid, their bending and branching in order to increase heat transfer area, and their topology may have a major impact on thermal performance and power consumption. In most cases, the design of heat exchangers is restricted to simple geometries, such as parallel-, cross- and counter-flow devices, which are easily assembled using standard components through

Responsible Editor: Kurt Maute

Topical Collection: Flow-driven Multiphysics
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conventional manufacturing processes (e.g., bending of tubes). Their design usually starts by selecting the type of the heat exchanger and, then, searching for the optimal size, length and number of tubes, configuration of fins, etc, Shah and Sekulic (2003); Nitsche and Gbadamosi (2016).

In recent years, there is an increasing demand for compact, unconventionally shaped heat exchangers with high thermal efficiency and low energy consumption. To create innovative designs, additive manufacturing techniques, such as Selective Laser Melting, Yap et al. (2015), or Electron Beam Melting, Zah and Lutzmann (2010); Murr et al. (2012), in which metal powder is placed in a step-by-step manner to create structures, can be used, Gibson et al. (2021). In addition, Computational Fluid Dynamics (CFD) and Conjugate Heat Transfer (CHT) methods and software, simulating mass flow and heat transfer phenomena, can numerically predict the performance of heat exchangers. Incorporating them into optimization workflows can lead to new better performing devices. For instance, in Hilbert et al. (2006), a two-objective optimization of the fin-blade shapes of a tubular heat exchanger was carried out using a genetic algorithm. Half of the symmetric blade was parameterized using NURBS, with 4 control points; the position of blades inside the heat exchanger was not allowed to change. In Najafi et al. (2011), a genetic algorithm was also used to design a plate-and-fin heat exchanger with high thermal efficiency and low manufacturing and maintenance cost. However, in cases of highly complex and narrow geometries, the ideal topology/connectivity of fluid paths might not be obvious. TopO can be used in such problems to fully profit of the design freedom offered by additive manufacturing.

TopO is a material distribution method, originally proposed by Bendsoe and Kikuchi (1988) for solid mechanics and structures, that computes the optimal material layout over a design domain with respect to (w.r.t.) certain performance metrics and design constraints. In fluid mechanics, there are various TopO formulations, including phase-field approaches, Bourdin and Chambolle (2003), level-set methods, Makhija and Maute (2015); Zhang and Liu (2015); Kubo et al. (2017); Sasaki et al. (2019), porosity/density-based methods, Borrvall and Petersson (2003); Gersborg-Hansen et al. (2005), etc. The latter, used in the present article, makes use of a continuous porosity/permeability field to distinguish regions occupied by fluid or solid. The artificially introduced porosity-like terms act as blockage terms into the underlying equations driving flow velocity towards zero in solidified areas. TopO has a vast design space, with as many design variables as the number of grid cells, which makes it ideal for designing optimal fluid paths/manifolds when there is no a-priori knowledge of the connectivity between the inlets and outlets. In fluid mechanics, TopO has been applied to a wide range of problems, including Stokes flows, Borrvall and Petersson (2003), laminar, Gersborg-Hansen et al. (2005), and turbulent flows, Yoon (2016); Papoutsis-Kiachagias and Giannakoglou (2016); Dilgen et al. (2018), unsteady flows, Deng et al. (2011); Nørgaard et al. (2016), reactive flows, Okkels and Bruus (2007), natural convection problems, Alexandersen et al. (2014, 2016), etc. The interested reader can find an overview of state-of-the-art TopO methods for fluid flow problems, including heat transfer, in a recent article by Alexandersen and Andreasen (2020). The use of TopO for fluid problems with CHT first appeared in Dede (2009); Yoon (2010) and still remains an active area of research. TopO has been used only recently in order to design bi-fluid heat exchangers. In Kobayashi et al. (2019), TopO is used to create guiding winglets increasing heat transfer in a fin-and-tube heat exchanger, whereas in Høghøj et al. (2020); Kobayashi et al. (2021) TopO is used to generate optimal heat exchanger geometries based only on the positions of inlets/outlets for the two fluids.

To reduce the computational cost of performing a CHT TopO based on the full-set of non-linear Reynolds-Averaged Navier-Stokes (RANS) equations in fluid problems, a simplified flow model, resembling Darcy's law in porous media, has been proposed. This model is derived by neglecting inertial terms and viscous effects from the Navier-Stokes equations. In Zhao et al. (2018), the Darcy-like model was used to perform density-based TopO for cooling channel applications. Then, it was employed for 2D, Asmussen et al. (2019), and 3D, Pollini et al. (2020), TopO of natural convection problems with applications in heat-sinks design. Due to the introduced simplifications, the cost of TopO based on the Darcy-like model is significantly lower than that of a RANS-TopO. However, the obtained Darcy-TopO solutions should be re-evaluated with a full Navier-Stokes solver in order to verify their performance, Pollini et al. (2020).

Despite its huge upsides, TopO's main shortcoming is the lack of an accurate representation of the FSI since boundary conditions are imposed in a weak sense through blockage terms in the flow equations. Unfortunately, this allows some flow leakage through the solidified parts of the domain and makes TopO prone to inaccuracies, occasionally leading to sub-optimal solutions. Quite often, when the TopO solutions are re-evaluated using a grid fitted to the computed FSI, they do not perform as expected. This provided an incentive for successively running TopO and ShpO, Koch et al. (2017). ShpO, Pironneau (1983), requires either a shape parameterization/morphing technique or the direct control of boundary nodes, and such a decision affects the sought optimal solution. ShpO is not as versatile as TopO in the sense that it cannot account for topological changes in the initial design, e.g., the creation of new fluid paths and/or the blockage of existing ones. However, its great advantage is that boundary conditions are imposed in the strong sense.

In this paper, a continuous adjoint-based framework that combines a) the versatility of TopO to generate optimal flow

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paths when there is no obvious way to connect inlets and outlets and b) the imposition of accurate boundary conditions in ShpO, both for bi-fluid CHT problems, is presented and used to design a compact heat exchanger, with many inlets and outlets. First, a bi-fluid CHT TopO is used to get close to the optimal design, targeting max. heat transfer and min. total pressure losses for each fluid stream. Constraints on the volume occupied by each fluid and the outlet flowrates balance for one of the fluids are imposed. An adjoint-based Darcy-like (low-fidelity) TopO is used to initialize the porosity field at lowcost, as in Zhao et al. (2018); Pollini et al. (2020), without though including any parameter tuning for the permeability of the fluid, before continuing with the RANS-based (high-fidelity) TopO. Then, Topology-to-Shape Transition (TtoST) follows; this transforms the separators of fluid/solid cells into a smooth surface grid and generates appropriate 3D grids in the three domains of interest. The CHT ShpO starts from the geometry generated by the TtoST, and its purpose is to further refine the TopO solution. To compute gradients, both TopO and ShpO make use of the continuous adjoint method, Jameson (1988); Anderson and Venkatakrishnan (1999); Papoutsis-Kiachagias and Giannakoglou (2016), which also includes the adjoint to the Spalart–Allmaras one-equation turbulence model, Zymaris et al. (2009). Everything is implemented within the open-source CFD toolbox of OpenFOAM©.

2 Objective and constraint functions

The synergistic use of TopO and ShpO for the design of heat exchangers operating with a cold ($\gamma = 1$; coolant) and hot ($\gamma = 2$; cooled) fluid is based on the same objective and constraint functions. Using different sets of design variables (generally denoted by $\mathbf{b}$), TopO and ShpO aim at minimizing an objective function J, subject to m inequality constraints expressed by g_i, namely

$$\begin{aligned} &\min_{\mathbf{b}} J(\mathbf{b}, \mathbf{U}(\mathbf{b})) \\ &s.t. \quad g_i(\mathbf{b}, \mathbf{U}(\mathbf{b})) \leq 0\,, \quad i = 1, ..., m \end{aligned} \tag{1}$$

where $\mathbf{U} = \left[p^{(\gamma)}, v_i^{(\gamma)}, \widetilde{\nu}^{(\gamma)}, \Delta, T\right] (\gamma = 1, 2)$ is the array of flow variables, $p^{(\gamma)}$ is the static pressure divided by the fluid density, $v_i^{(\gamma)}$ are Cartesian velocity components, $\widetilde{\nu}^{(\gamma)}$ is the turbulence model variable and Δ denotes the distance to the nearest solid wall. T is the temperature over both fluids and the solid.

The heat fluxes of the cold and hot fluids, measured at their outlets $S_O^{(1)}, S_O^{(2)}$, are defined as

$$J_E^{(\gamma)} = \int_{S_O^{(\gamma)}} \rho^{(\gamma)} c_p^{(\gamma)} v_j^{(\gamma)} n_j T dS \tag{2}$$

where n_j are the components of the normal unit vector pointing outward of the fluid domains and $\rho^{(\gamma)}$ and $c_p^{(\gamma)}$ are the densities and specific heat capacities of the two fluids. Repeated indices imply summation, according to the Einstein convention. In the design of heat exchangers, the main target is to maximize the amount of heat transferred between the coolant ($\gamma = 1$) and the cooled fluid ($\gamma = 2$). The conservation of energy implies that this is equivalent to the minimization of $J_E^{(2)}$, to be used as the first objective function.

Additionally, the total energy consumption for operating the heat exchanger, which is equivalent to the (volume flowrate weighted) total pressure ($p_t^{(\gamma)} = p^{(\gamma)} + \frac{1}{2} v_k^{(\gamma)^2}$) drop for each fluid between its inlets and outlets, should be kept as low as possible. Therefore, a pair ($\gamma = 1, 2$) of additional objectives,

$$J_P^{(\gamma)} = -\int_{S_I^{(\gamma)}} v_j^{(\gamma)} n_j p_t^{(\gamma)} dS - \int_{S_O^{(\gamma)}} v_j^{(\gamma)} n_j p_t^{(\gamma)} dS \tag{3}$$

(both to be minimized) must be considered. Herein, a single-objective optimization problem is solved, minimizing

$$J = w_E \frac{J_E^{(2)}}{J_{E,0}^{(2)}} + w_P \sum_{\gamma=1,2} \frac{J_P^{(\gamma)}}{J_{P,0}^{(\gamma)}} \tag{4}$$

where w_E, w_P are weights defined by the designer and $J_{E,0}^{(2)}$, $J_{P,0}^{(\gamma)}$ denote the objective values at the beginning of TopO or ShpO loop.

A typical constraint related to fluid streams exiting from many outlets (herein, the cold stream, $\gamma = 1$, in Fig. 2) is to control the volume flowrate ratios Q_{O_k}/Q_I, by satisfying

$$g_M = \frac{1}{2} \sum_{k=1}^{K} \left(\frac{Q_{O_k}}{Q_I} - r_k \right)^2 - \epsilon \leq 0 \tag{5}$$

where K is the number of exits, $Q_{O_k} = \int_{S_{O_k}^{(1)}} v_j^{(1)} n_j dS$ is the volume flowrate at the k–th exit, $Q_I = -\int_{S_I^{(1)}} v_j^{(1)} n_j dS$ is the fixed incoming flowrate, ϵ is a small threshold value and r_k are percentages defined by the designer so that $\sum_{k=1}^{K} r_k = 1$. If all outgoing flowrates should be balanced, as in this paper, then $r_k = \frac{1}{K}, \forall k$.

Finally, an additional constraint is related to the volume of the domain occupied by each fluid, namely

$$g_V^{(\gamma)} = \frac{\int_\Omega \xi^{(\gamma)} d\Omega}{\int_\Omega d\Omega} - V_{target}^{(\gamma)} \leq 0\,, \quad \gamma = 1, 2 \tag{6}$$

where Ω is the design domain, $V_{target}^{(\gamma)}$ is the maximum allowed volume percentage for fluid γ. In Eq. 6, $\xi^{(\gamma)}$ ($\gamma = 1, 2$) stand for two auxiliary indicator fields, with $\xi^{(\gamma)} = 1$ in areas occupied by fluid γ and 0 elsewhere. Hence, $\xi^{(1)} = \xi^{(2)} = 0$

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denotes the solidified parts of Ω. The constraint of Eq. 6 also facilitates the solidification of isolated fluid pockets that can be formed during TopO.

3 TopO for Bi-fluid applications

A single porosity field α is used to identify parts in Ω occupied by solid or any of the fluids, with values varying continuously between 0 and 1. $\alpha = 0$ corresponds to areas where the cold fluid ($\gamma = 1$) flows, $\alpha = 1$ corresponds to the hot fluid ($\gamma = 2$) whereas areas occupied by solid material ideally have $\alpha = 0.5$; areas with $0 < \alpha < 0.5$ or $0.5 < \alpha < 1$ correspond to "gray" regions between solid and fluid. TopO seeks to find the optimal α field (design variables) to minimize the objective function(s) by meeting the constraints mentioned in Sect. 2.

3.1 Governing equations

For each topological solution with the current $\xi^{(\gamma)}$ fields, the mean-flow equations for incompressible fluids, for $\gamma = 1, 2$, coupled with the Spalart–Allmaras turbulence model Spalart and Allmaras (1992) and the Hamilton-Jacobi equation, Tucker (2003), that computes the necessary distances Δ from the fluid–solid interfaces (solid walls), to be solved over the entire domain Ω, are

$$R^{p^{(\gamma)}} = -\frac{\partial v_j^{(\gamma)}}{\partial x_j} = 0 \qquad \text{湍流控制方程} \tag{7a}$$

$$R_i^{v^{(\gamma)}} = v_j^{(\gamma)}\frac{\partial v_i^{(\gamma)}}{\partial x_j} - \frac{\partial \tau_{ij}^{(\gamma)}}{\partial x_j} + \frac{\partial p^{(\gamma)}}{\partial x_i} + \beta_{max} I\left(\xi^{(\gamma)}\right) v_i^{(\gamma)} = 0\,, \quad i = 1, 2, 3 \tag{7b}$$

$$R^{\tilde{\nu}^{(\gamma)}} = v_j^{(\gamma)}\frac{\partial \tilde{\nu}^{(\gamma)}}{\partial x_j} - \frac{\partial}{\partial x_j}\left[\left(\nu^{(\gamma)} + \frac{\tilde{\nu}^{(\gamma)}}{\sigma}\right)\frac{\partial \tilde{\nu}^{(\gamma)}}{\partial x_j}\right] - \frac{c_{b2}}{\sigma}\left(\frac{\partial \tilde{\nu}^{(\gamma)}}{\partial x_j}\right)^2 - \tilde{\nu}^{(\gamma)} P\left(\tilde{\nu}^{(\gamma)}, \Delta\right) + \tilde{\nu}^{(\gamma)} D\left(\tilde{\nu}^{(\gamma)}, \Delta\right) + \beta_{max} I\left(\xi^{(\gamma)}\right)\tilde{\nu}^{(\gamma)} = 0 \tag{7c}$$

$$R^{\Delta} = \frac{\partial}{\partial x_j}\left(\frac{\partial \Delta}{\partial x_j}\Delta\right) - (1+\epsilon)\Delta\frac{\partial^2 \Delta}{\partial x_j^2} - 1 + \beta_{max} I\left(\xi^{(1)} + \xi^{(2)}\right)\Delta = 0 \tag{7d}$$

In Eqs. 7, $\tau_{ij}^{(\gamma)} = \left(\nu^{(\gamma)} + \nu_t^{(\gamma)}\right)\left(\frac{\partial v_i^{(\gamma)}}{\partial x_j} + \frac{\partial v_j^{(\gamma)}}{\partial x_i}\right)$ is the stress tensor, $\nu^{(\gamma)}$ and $\nu_t^{(\gamma)} = f_{v1}^{(\gamma)}\tilde{\nu}^{(\gamma)}$ are the bulk and eddy viscosities. $D\left(\tilde{\nu}^{(\gamma)}, \Delta\right)$ and $P\left(\tilde{\nu}^{(\gamma)}, \Delta\right)$ are the turbulence dissipation and production terms, respectively, and require the distance Δ from the nearest wall boundaries to be computed. c_{b2}, σ are constants (see Appendix A).

Equations 7c, 7d are coupled through the appearance of Δ in the production and dissipation terms of the turbulence model. Equation 7d is written in a convective form by, also, including some additional diffusion; the latter leads to smoother distance fields. In case the diffusion coefficient ϵ is set to 0, Eq. 7d corresponds to the typical Hamilton-Jacobi equation. However, non-zero values are preferred to ensure that the solution of Eq. 7d is numerically stable. Here, $\epsilon = 0.1$. Note that, Eq. 7d is solved once over Ω for a single Δ field.

The last terms in Eqs. 7b, 7c, 7d correspond to the Brinkman penalization terms, Borrvall and Petersson (2003), that drive the corresponding variable field to zero over solidified areas and, in regard to Eqs. 7b, 7c, over areas in which the other fluid is flowing. β_{max} stands for the impermeability of the hypothetical porous media in areas where fluid γ should not flow; theoretically, $\beta_{max} \to \infty$. In practice, its value is computed based on Darcy's law, Olesen et al. (2006), expressing the ratio between viscous and porous forces, as follows

$$\beta_{max} = \frac{\nu}{Da \cdot \mathcal{L}^2} \tag{8}$$

where Da is the Darcy coefficient ($Da \approx 10^{-5}$) and $\mathcal{L}$ the characteristic length-scale of the problem (here, $\mathcal{L} = 5mm$, corresponding to the side length of the inlet/outlet ducts of Fig. 2). The lower the Da value, the higher the impermeability. $I\left(\xi^{(\gamma)}\right)$ stands for a sharpening function, Borrvall and Petersson (2003), defined by

$$I\left(\xi^{(\gamma)}\right) = \frac{1 - \xi^{(\gamma)}}{1 + b_I \xi^{(\gamma)}} \tag{9}$$

with b_I controlling the convexity of the function.

Heat transfer over Ω is modeled through a single convection–diffusion equation which is solved for T. In incompressible fluid flows, the solution of the flow equations affects heat transfer but not the other way around; hence, Eqs. 7 are solved before (i.e., are decoupled from) the energy equation. Since $v_i^{(\gamma)}$ is assumed to be zero outside the region occupied by fluid γ, the energy equation is written once, for the entire domain Ω, in the form

温度控制方程

$$R^T = \sum_{\gamma=1,2} \rho^{(\gamma)} c_p^{(\gamma)} v_j^{(\gamma)}\frac{\partial T}{\partial x_j} - \frac{\partial}{\partial x_j}\left(k_{eff}\frac{\partial T}{\partial x_j}\right) = 0 \tag{10}$$

where k_{eff} is the effective thermal conductivity. The discrete form of Eq. 10 using finite volumes requires k_{eff} only at the cell faces. A linear interpolation between the cell values neighboring each face can lead to inaccuracies at the faces separating fluid and solid cells; in cases where $k_{eff}^{(\gamma)} = k^{(\gamma)} + \frac{\rho^{(\gamma)} c_p^{(\gamma)} \nu_t^{(\gamma)}}{Pr_t^{(\gamma)}}$ for fluid γ (with $k^{(\gamma)}$ being its thermal conductivity and Pr_t the turbulent Prandtl number) and k_S (thermal conductivity of the solid) differ by orders of magnitude, this may erroneously compute heat transfer at the

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FSI. To this end, the following interpolation is proposed to compute the k_{eff} values at the grid faces, (superscript f)

$$\begin{aligned} k_{eff}^{f} &= H(\xi^{(1,f)}, b_H, \eta)k_{eff}^{(1,f)} + H(\xi^{(2,f)}, b_H, \eta)k_{eff}^{(2,f)} \\ &+ H(1-\xi^{(1,f)}-\xi^{(2,f)}, b_H, \eta)k_S \\ &+ \frac{k_N k_O \mathcal{D}^{NO}}{k_N \mathcal{D}^{Of} + k_O \mathcal{D}^{Nf}}\left[\Phi(\xi^{(1,f)}, b_\Phi) + \Phi(\xi^{(2,f)}, b_\Phi)\right] \end{aligned} \tag{11}$$

where $\xi^{(\gamma,f)}$ ($\gamma = 1, 2$) are the face values of the indicator fields (theoretically, $\xi^{(\gamma)} = 0.5$ at a face separating solid and fluid γ), linearly interpolated from the grid cells N and O surrounding it. $\mathcal{D}^{NO}$, $\mathcal{D}^{Of}$ and $\mathcal{D}^{Nf}$ are the distances between N, O and f, as indicated by the exponents, k_N and k_O are the values of k at the N and O cells, respectively, given by

$$k\left(\xi^{(1)}, \xi^{(2)}\right) = k^{(1)}\xi^{(1)} + k^{(2)}\xi^{(2)} + k_S\left(1-\xi^{(1)}-\xi^{(2)}\right) \tag{12}$$

The Φ and H functions are given by

$$\Phi\left(\xi^{(\gamma)}, b_\Phi\right) = \left[4\xi^{(\gamma)}(1-\xi^{(\gamma)})\right]^{b_\Phi} \tag{13}$$

$$H\left(\xi^{(\gamma)}, b_H, \eta\right) = \frac{tanh\left(\eta b_H\right) + tanh\left(b_H(\xi^{(\gamma)}-\eta)\right)}{tanh\left(\eta b_H\right) + tanh\left(b_H(1-\eta)\right)} \tag{14}$$

where b_Φ is a parameter increasing the steepness of Φ close to $\xi^{(\gamma)} = 0.5$, η is a threshold value (equal to 0.95 for use in Eq. 11) and b_H is a steepening parameter (see also Sect. 3.2). The first three terms on the right-hand-side (RHS) of Eq. 11 correspond to the linear interpolation of k_{eff} based on its values at the neighboring cells, in case these cells are occupied by the same material (i.e., if one of $\xi^{(1,f)}$, $\xi^{(2,f)}$ or $1-\xi^{(1,f)}-\xi^{(2,f)}$ is approximately equal to 1). The last term on the RHS of Eq. 11 is introduced to correct this value by making it consistent with that computed at the FSI in a simulation using body-fitted grids (see derivation in Appendix B).

Regarding the boundary conditions for Eq. 10, Dirichlet conditions are imposed on T at the inlets and zero Neumann conditions at the outlets, for both fluids. The remaining parts of the boundary S of Ω are considered to be thermally insulated, where $\frac{\partial T}{\partial n} = 0$.

3.2 Processing the porosity field

Using the α field (as this is updated in each optimization cycle) directly in the flow equations may lead to TopO solutions which exhibit checkerboard-like patterns, Bruns and Tortorelli (2001). To ensure smoothly varying porosity fields, filtering schemes are usually employed; these also mitigate the effect of the grid to the TopO solution, Bourdin and Chambolle (2003). Given the α field, the numerical solution of a Helmholtz-type PDE Lazarov and Sigmund (2011); Lazarov et al. (2016),

$$R^F(\widetilde{\alpha}, \alpha) = -\left(\frac{r}{2\sqrt{3}}\right)^2 \frac{\partial^2 \widetilde{\alpha}}{\partial x_j^2} + \widetilde{\alpha} - \alpha = 0 \tag{15}$$

where r is the smoothing radius, provides a new smoothly varying porosity field $\widetilde{\alpha}$. Zero Neumann conditions are imposed on $\widetilde{\alpha}$ along S.

As Eq. 15 may considerably smooth out the porosity field, wide "gray" regions between the fluid and the solid may appear. To get a more "binary" representation of the interface, after having computed the $\widetilde{\alpha}$ field, a projection step follows to steepen it; this is based on a smooth Heaviside function (Eq. 14), Wang et al. (2011), computing β at each cell as

$$\beta = H\left(\widetilde{\alpha}, b_H, \eta\right) \tag{16}$$

where $\eta = 0.5$. Eq. 15, 16 need to be applied twice to ensure a minimum thickness t_e of the wall separating the two fluids, Clausen et al. (2015); Høghøj et al. (2020). The second filtering applies to β, using a different (erosion) radius r_e, yielding field $\widetilde{\beta}$. Then, $\widetilde{\beta}$ is projected twice, using a low $\eta^{(1)} = 0.05$ and a high $\eta^{(2)} = 0.95$ threshold value, yielding the $\widetilde{\xi}^{(1)}$ and $\widetilde{\xi}^{(2)}$ fields, respectively. Grid cells are marked as fluid 1, fluid 2 or solid according to the following rule

$$\begin{aligned} &Fluid\ 1 : \xi^{(1)} = 1 - \widetilde{\xi}^{(1)} = 1 \\ &Fluid\ 2 : \xi^{(2)} = \widetilde{\xi}^{(2)} = 1 \\ &Solid\ \ \ : \xi^{(1)} = \xi^{(2)} = 0 \end{aligned} \tag{17}$$

giving rise to the aforementioned $\xi^{(\gamma)}$ ($\gamma = 1, 2$) indicator fields. According to Luo et al. (2019), to obtain a wall thickness t_e, $r_e \approx 0.75 t_e$.

Alternatively, the projection and filtering steps required to compute the $\widetilde{\beta}$ field from $\widetilde{\alpha}$ can be replaced by the sum of two Heaviside functions, with a low η_L and a high η_H threshold value, respectively, in order to obtain areas with sharp values $\widetilde{\beta} = \{0, 1\}$ for the two fluids and $\widetilde{\beta} = 0.5$ for the solid. This function, which will be referred to as bi-projection and is plotted in Fig. 1, takes the following form

$$\widetilde{\beta} = BP(\widetilde{\alpha}) = \frac{1}{2}\left(H\left(\widetilde{\alpha}, b_H, \eta_L\right) + H\left(\widetilde{\alpha}, b_H, \eta_H\right)\right) \tag{18}$$

where b_H has the same role as in Eq. 16. Based on experience, $\eta_L = 0.3$, $\eta_H = 0.7$ and $b_H = 10$ are adequate to ensure that the two fluids are separated by solid material of adequate thickness.

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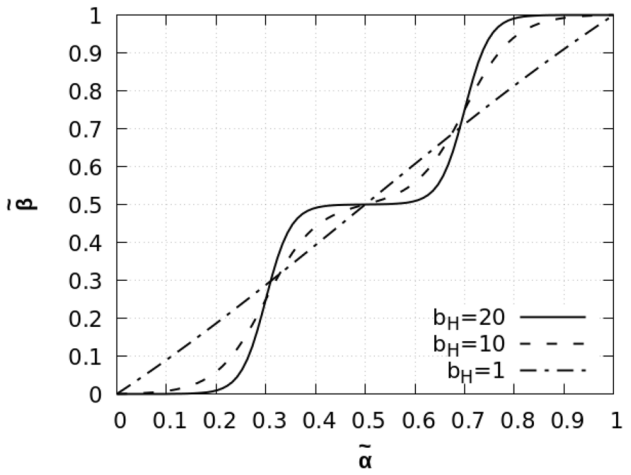


Fig. 1 The bi-projection function of Eq. 18 plotted for different steepening values b_H for threshold values of $\eta_L = 0.3$, $\eta_H = 0.7$.

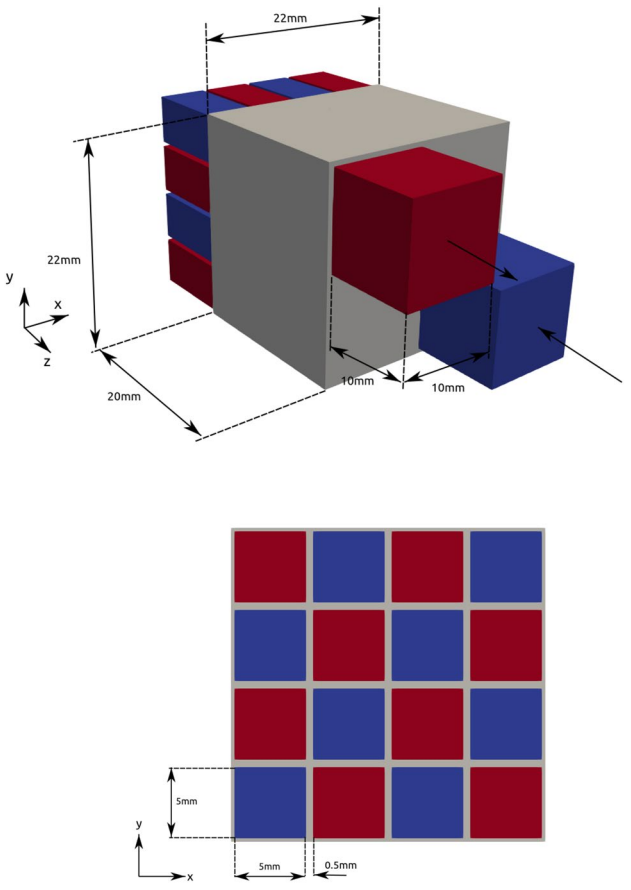


Fig. 2 Top: perspective view of the heat exchanger. Bottom: view of its many inlet/outlet side. The hot and cold streams are marked in red and blue, respectively, with the solid material marked in gray

3.3 Continuous adjoint method

The continuous adjoint method is used to compute $\frac{\delta J}{\delta \alpha}$, where J refers to either the objective or the constraint function. J is augmented with the field integrals of the residuals of the primal equations (Eqs. 7, 10) multiplied by the adjoint variable fields, giving rise to the Lagrangian

$$L = J + \sum_{\gamma=1,2} \int_{\Omega} \left(q^{(\gamma)} R^{p^{(\gamma)}} + u_i^{(\gamma)} R_i^{v^{(\gamma)}} + \widetilde{\nu}_a^{(\gamma)} R^{\widetilde{\nu}^{(\gamma)}} + \Delta_\alpha^{(\gamma)} R^{\Delta} \right) d\Omega + \int_{\Omega} T_\alpha R^T d\Omega \tag{19}$$

where q is the adjoint pressure, u_i are the components of the adjoint velocity, $\widetilde{\nu}_a$ the adjoint turbulence model variable, T_α the adjoint temperature and Δ_α the adjoint distance. After a lengthy mathematical development similar to that presented in Kontoleontos et al. (2013), (therein, though, for a single fluid problem), the variation of L w.r.t. the porosity value (α_m) at cell m takes the following form

$$\begin{aligned}
\frac{\delta L}{\delta \alpha_m} &= \sum_{\gamma=1,2} \int_{\Omega} \left(R^{q^{(\gamma)}} \frac{\partial p^{(\gamma)}}{\partial \alpha_m} + R_i^{u_i^{(\gamma)}} \frac{\partial v_i^{(\gamma)}}{\partial \alpha_m} + R^{\widetilde{\nu}_a^{(\gamma)}} \frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial \alpha_m} + R^{\Delta_\alpha^{(\gamma)}} \frac{\partial \Delta}{\partial \alpha_m} \right) d\Omega \\
&+ \sum_{\gamma=1,2} \int_{S} \left(BC_1 \frac{\partial p^{(\gamma)}}{\partial \alpha_m} + BC_{2,i} \frac{\partial v_i^{(\gamma)}}{\partial \alpha_m} + BC_3 \frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial \alpha_m} + BC_4 \frac{\partial \Delta}{\partial \alpha_m} \right) dS \\
&+ \int_{\Omega} R^{T_\alpha} \frac{\partial T}{\partial \alpha_m} d\Omega + \int_{S} BC_5 \frac{\partial T}{\partial \alpha_m} dS - \int_{S} T_\alpha \frac{\partial}{\partial \alpha_m} \left(k_{eff} \frac{\partial T}{\partial n} \right) dS \\
&- \sum_{\gamma=1,2} \int_{S} \left(\nu + \frac{\widetilde{\nu}^{(\gamma)}}{\sigma} \right) \widetilde{\nu}_a^{(\gamma)} \frac{\partial}{\partial \alpha_m} \left(\frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial n} \right) dS - \sum_{\gamma=1,2} \int_{S} u_i^{(\gamma)} n_j \frac{\partial \tau_{ij}^{(\gamma)}}{\partial \alpha_m} dS \\
&- \sum_{\gamma=1,2} \int_{S} \epsilon \Delta \Delta_\alpha^{(\gamma)} \frac{\partial}{\partial \alpha_m} \left(\frac{\partial \Delta}{\partial n} \right) dS + \int_{\Omega} \Theta_{\widetilde{\beta}} \frac{\partial \widetilde{\beta}}{\partial \beta} \frac{\partial \beta}{\partial \widetilde{\alpha}} \frac{\partial \widetilde{\alpha}}{\partial \alpha_m} d\Omega
\end{aligned} \tag{20}$$

where

$$\begin{aligned}
\Theta_{\widetilde{\beta}} = &\sum_{\gamma=1,2} \beta_{max} \left(v_i^{(\gamma)} u_i^{(\gamma)} + \widetilde{\nu}^{(\gamma)} \widetilde{\nu}_a^{(\gamma)} + \Delta \Delta_\alpha^{(\gamma)} \right) \frac{\partial I}{\partial \xi^{(\gamma)}} \frac{\partial \xi^{(\gamma)}}{\partial \widetilde{\beta}} \\
&+ \sum_{\gamma=1,2} \frac{\partial T_\alpha}{\partial x_j} \frac{\partial T}{\partial x_j} \left[\frac{\partial k}{\partial \xi^{(\gamma)}} + \frac{\nu_t}{Pr_t} \frac{\partial (\rho c_p)}{\partial \xi^{(\gamma)}} - \frac{\rho c_p \nu_t}{Pr_t^2} \frac{\partial Pr_t}{\partial \xi^{(\gamma)}} + \frac{\rho c_p}{Pr_t} \frac{\partial \nu_t}{\partial \xi^{(\gamma)}} \right] \frac{\partial \xi^{(\gamma)}}{\partial \widetilde{\beta}}
\end{aligned} \tag{21}$$

In order to make $\frac{\delta L}{\delta \alpha_m}$ independent of field integrals including the partial derivatives of $p^{(\gamma)}$, $v_i^{(\gamma)}$, $\widetilde{\nu}^{(\gamma)}$, Δ and T w.r.t. α_m,

$$R^{q^{(\gamma)}} = R_i^{u_i^{(\gamma)}} = R^{\widetilde{\nu}_a^{(\gamma)}} = R^{\Delta_\alpha^{(\gamma)}} = R^{T_\alpha} = 0 \tag{22}$$

should hold, giving rise to the adjoint field equations, summarized in Appendix C.

Making $\frac{\delta L}{\delta \alpha_m}$ independent of the same derivatives along S results to the adjoint boundary conditions (see

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Papoutsis-Kiachagias and Giannakoglou (2016); Kontoleontos et al. (2013)).

Having satisfied the field adjoint equations and boundary conditions, the remaining terms in Eq. 20 stand for the derivatives of J w.r.t. α_m; thus, for cell m

$$\begin{aligned}\frac{\delta J}{\delta \alpha_m} &= \int_\Omega \Theta_{\widetilde{\beta}} \frac{\partial \widetilde{\beta}}{\partial \beta}\frac{\partial \beta}{\partial \widetilde{\alpha}}\frac{\partial \widetilde{\alpha}}{\partial \alpha_m} d\Omega \\ &- \int_\Omega \Psi_{\widetilde{\beta}} \frac{\partial R^F(\widetilde{\beta}, \beta)}{\partial \alpha_m} d\Omega \\ &- \int_\Omega \Psi_{\widetilde{\alpha}} \frac{\partial R^F(\widetilde{\alpha}, \alpha)}{\partial \alpha_m} d\Omega\end{aligned} \tag{23}$$

In Eqs. 21 and 23, $\frac{\partial \xi^{(\gamma)}}{\partial \widetilde{\beta}}$ and $\frac{\partial \beta}{\partial \widetilde{\alpha}}$ are computed analytically by differentiating the Heaviside projection function (Eq. 16). Eq. 23 has purposefully been augmented by the last two integrals to avoid the costly computation of $\frac{\partial \widetilde{\beta}}{\partial \beta}$ and $\frac{\partial \widetilde{\alpha}}{\partial \alpha_m}$. Through a similar development as the one used to derive the field adjoint equations, the adjoint to the Helmholtz filtering PDE (Eq. 15) arises in the following form

$$R^{\Psi_\zeta} = -\left(\frac{r}{2\sqrt{3}}\right)^2 \frac{\partial^2 \Psi_\zeta}{\partial x_j^2} + \Psi_\zeta - \Theta_\zeta = 0 \tag{24}$$

where index ζ denotes either $\widetilde{\beta}$ or $\widetilde{\alpha}$, whereas Ψ_ζ are additional adjoint variables. Eq. 24 needs to be solved twice, first for $\zeta = \widetilde{\beta}$, with $\Theta_{\widetilde{\beta}}$ given by Eq. 21, and then for $\zeta = \widetilde{\alpha}$, with $\Theta_{\widetilde{\alpha}} = \Psi_{\widetilde{\beta}} \frac{\partial \beta}{\partial \widetilde{\alpha}}$. The final expression of the derivative of J w.r.t. α_m is

$$\frac{\delta J}{\delta \alpha_m} = \Psi_{\widetilde{\alpha}}^m \Omega^m \tag{25}$$

where Ω^m is the volume of the m-th cell; repeated superscripts on the RHS of Eq. 25 do not imply summation.

3.4 Initialization of the porosity field in Bi-fluid TopO problems

Starting TopO with a sensible initialization for the α field may considerably reduce its cost. In single fluid TopO problems, an all-fluid initialization is often used, since the volume constraints of Eq. 6 can solidify isolated fluid pockets that might emerge during TopO. However, in bi-fluid TopO, such an initialization is not appropriate since using $\alpha = 0$ or $\alpha = 1$ everywhere in Ω will completely block the design domain for one of the fluids and might bias the TopO solution towards the other. Initializing Ω with $\alpha = 0.5$ (i.e., solid) may cause TopO to run for a large number of optimization cycles until paths for both fluids are carved into the solidified domain.

Herein, prior to performing the CHT TopO using the flow model presented in Sect. 3.1, a much cheaper TopO run, to be seen as an extension of the work of Zhao et al. (2018); Pollini et al. (2020), is used. This computes the α field from which the (high-fidelity) RANS-based CHT TopO of Sect. 3.1 starts.

The low-fidelity Darcy-like model can be derived from Eq. 7b, by neglecting convective terms, diffusion by turbulence and the turbulence model itself and assuming that the divergence of the stress tensor depends linearly on the corresponding velocity vector, i.e.,

$$\frac{\partial \tau_{ij}^{(\gamma)}}{\partial x_j} = -\frac{\nu^{(\gamma)}}{\mathcal{L}^2} v_i^{(\gamma)} \tag{26}$$

where $\mathcal{L}$ is the same characteristic length used in Eq. 8. With these assumptions, Eq. 7b becomes

$$v_i^{(\gamma)} = -\frac{1}{\beta_{max}\left(Da + I\left(\xi^{(\gamma)}\right)\right)} \frac{\partial p^{(\gamma)}}{\partial x_i} \tag{27}$$

according to which the velocity is proportional to the pressure gradient.

Substituting Eq. 27 into the continuity equation (Eq. 7a) yields

$$\frac{\partial}{\partial x_j}\left(\frac{1}{\beta_{max}\left(Da + I\left(\xi^{(\gamma)}\right)\right)} \frac{\partial p^{(\gamma)}}{\partial x_j}\right) = 0\,, \quad \gamma = 1, 2 \tag{28}$$

which gives two Laplace equations with spatially varying coefficients, to be solved for $p^{(\gamma)}$. Their boundary conditions are: at S_I, $\frac{\partial p^{(\gamma)}}{\partial n} = -\frac{\nu}{\mathcal{L}^2} v_i^{(\gamma)} n_i$; at S_O, $p^{(\gamma)} = 0$; at S_W, $\frac{\partial p^{(\gamma)}}{\partial n} = 0$. The solution of this simplified flow model can be obtained at a small fraction of the time needed for the solution of the Navier-Stokes equations (around 3 orders of magnitude lower, in the examined case).

Upon solving Eqs. 28 for both fluids ($\gamma = 1, 2$), velocities are computed from Eq. 27. These are substituted into the convection-diffusion energy equation (Eq. 10) which is solved for T, without taking the turbulent thermal conductivity into account, i.e., $k_{eff}^{(\gamma,f)} = k^{(\gamma)}$ in Eq. 11.

For the low-cost initialization of TopO, an objective function which the adjoint method must be adapted to, is needed. Since the inertial forces in Eq. 7b were eliminated in the low-fidelity flow model, the power consumption is given by the volume-averaged static pressure drop from the inlet to the outlet of each fluid, written as $-\int_{S_I^{(\gamma)}} v_i^{(\gamma)} n_i p^{(\gamma)} dS - \int_{S_O^{(\gamma)}} v_i^{(\gamma)} n_i p^{(\gamma)} dS$. This can also be expressed as the volume integral $-\int_\Omega v_i^{(\gamma)} \frac{\partial p^{(\gamma)}}{\partial x_i} d\Omega$. Thanks to Eq. 27, the latter leads to the minimization of

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$$J_{PL}^{(\gamma)} = \int_{\Omega} \left(Da + I\left(\xi^{(\gamma)}\right)\right) v_i^{(\gamma)^2} d\Omega \tag{29}$$

In other words, the Darcy-like TopO, relying upon the continuous adjoint method, derived for the simplified model similarly to Sect. 3.3, seeks to minimize Eq. 4 by replacing $J_P^{(\gamma)}$ with Eq. 29. The usual constraints are imposed regarding the outgoing flowrates at the many outlets (Eq. 5) and the volume occupied by each fluid (Eq. 6).

3.5 The TopO algorithm

The TopO problem, with the objective functions and constraints presented in Sect. 2, is solved using the Method of Moving Asymptotes (MMA), Svanberg (1987), implemented in OpenFOAM©. The latter can handle the inequality constraints of Sect. 2 as well as porosity bound constraints ($\alpha \in [0, 1]$) at all cells.

The steps of the TopO algorithm are:

1. Initialize the porosity field α over the entire domain Ω by iteratively performing steps 2–6 (see below) based on the low-fidelity Darcy-like model of Sect. 3.4 and its adjoint.
2. Process α, as described in Sect. 3.2, to obtain the indicator fields $\xi^{(1)}$ and $\xi^{(2)}$.
3. Solve the primal (Eqs. 7, 10) and adjoint (Eqs. C1) flow equations.
4. Compute derivatives for all objectives and constraints w.r.t. the α field, using Eqs. 21, 24 and 25.
5. Update the α field using the MMA method.
6. Return to step 2 or terminate TopO according to convergence criteria.

4 Transition from TopO solutions to body-fitted grids

The design obtained upon termination of the TopO process must be exported to a format suitable for the generation of a grid fitted to the FSI, in order to continue with ShpO. In the past, a way to extract the zero level-set surface corresponding to the FSI and represent it in an analytical CAD format based on splines has been reported for 2D TopO solutions by the group of authors, Koch et al. (2017). Herein, a different approach to export the FSI of the bi-fluid TopO solution into a CAD format is followed; this can be used either to proceed with manufacturing or continue with ShpO, by generating the necessary body-fitted grid.

The newly proposed TtoST includes the following three steps:

1. Cells with values of $\xi^{(\gamma)}$ greater than $\eta^{(\gamma)}$ (i.e., the threshold values used in the last projection step, see Sect. 3.2) are identified for $\gamma = 1, 2$. Based on the so-computed lists of cell indices, three subsets of the TopO grid, one for each fluid and one for the solid, are created.
2. The interfaces between these subsets correspond to the FSI boundary which is aligned with internal TopO grid faces resulting to a rough, staircase-like geometry. Center coordinates of these boundary faces (m_i) are then processed in order to acquire a smooth representation of the boundary. This requires the numerical solution of a Laplace-Beltrami equation, reading

$$R_i^{\widetilde{m}} = -r_m^2 \frac{\partial^2 \widetilde{m}_i}{\partial x_j^2} + \widetilde{m}_i - m_i = 0\,, \qquad i = 1, 2, 3 \tag{30}$$

   where $\widetilde{m}_i$ are the new smoothed face-center coordinates and r_m is the smoothing radius computed as the mean edge length of all FSI faces. The FSI grid nodes are, then, re-located by means of an inverse distance weighting interpolation of $\widetilde{m}_i$.
3. Boundary faces of the smoothed FSI are triangulated, if necessary, using face vertices and the corresponding face-center, and exported in STL file format (i.e., discrete/non-parameterized) which is compatible with most grid-generators, CAD toolboxes and 3D printers.

5 Bi-fluid CHT ShpO

The outcome of the TtoST process is a discrete representation of the FSIs of TopO. This is provided as input to a grid-generator yielding the 3D grids in the three areas of interest. The generated grids are used to evaluate anew the TopO optimized solutions by imposing exact boundary conditions along the FSI; by doing so, noticeable differences in the objective and constraint values are occasionally observed. This is mainly due to the inability of TopO to impose boundary conditions in the strong sense. Another reason is related to differences in the quality/density of grids used in TopO and ShpO: body-fitted grids are adequately refined close to the walls to accurately capture the flow and thermal boundary layers whereas TopO typically uses uniform Cartesian grids of lower resolution. This is why, ShpO should follow TopO, using the TtoST process to obtain the starting geometries.

In this paper, ShpO is performed by using volumetric B-Splines morphing lattices (the interested reader is referred to Papoutsis-Kiachagias et al. (2015)) to parameterize both the boundary and the interior nodes of the generated body-fitted grids. These lattices are also used as the mesh deformation tool during ShpO. The coordinates of selected

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control points of these morphing boxes act as the design variables b_l of the ShpO problem.

In CHT ShpO, Ω is decomposed into two fluid regions, denoted by $\Omega_F^{(\gamma)}, \gamma = 1, 2$, and the solid region Ω_S. The flow inside $\Omega_F^{(\gamma)}$ is governed by Eqs. 7, excluding the blockage terms of TopO. Heat transfer over the fluids is modeled by the convection-diffusion equations,

$$R^{T_F^{(\gamma)}} = \rho^{(\gamma)} c_p^{(\gamma)} v_j^{(\gamma)} \frac{\partial T^{F^{(\gamma)}}}{\partial x_j} - \frac{\partial}{\partial x_j}\left(k_{eff}^{(\gamma)} \frac{\partial T^{F^{(\gamma)}}}{\partial x_j}\right) = 0\,, \quad \gamma = 1, 2 \tag{31}$$

The boundary of each fluid domain $\Omega_F^{(\gamma)}$ comprises inlets $S_I^{(\gamma)}$, outlets $S_O^{(\gamma)}$, adiabatic solid walls $S_{F,W}^{(\gamma)}$ and FSI walls $\overline{S_F}^{(\gamma)}$.

Over the solid domain Ω_S, heat transfer is modeled via the heat conduction equation,

$$R_S^T = -\frac{\partial}{\partial x_j}\left(k_S \frac{\partial T^S}{\partial x_j}\right) = 0 \tag{32}$$

The boundary of Ω_S is $S_S = S_{S,W} \cup \overline{S_S}$ (same notation as for the fluids).

Boundary conditions for the energy equations on all boundaries except $\overline{S_F}^{(\gamma)}$ and $\overline{S_S}$ can be found in Sect. 3.1. Similarly to the TopO process, the heat transfer problem is solved after the solution of the mean-flow PDEs for each fluid. Eqs. 31 and 32 are solved simultaneously due to their coupling along the FSI boundaries ($\overline{S_F}^{(\gamma)}$ and $\overline{S_S}$), where temperature equality and heat flux conservation is imposed, i.e.,

$$T^{F^{(\gamma)}}|_{\overline{S_F}^{(\gamma)}} = T^S|_{\overline{S_S}}, \quad \underbrace{k_{eff}^{(\gamma)} \frac{\partial T^{F^{(\gamma)}}}{\partial n}\bigg|_{\overline{S_F}^{(\gamma)}}}_{Q^{F^{(\gamma)}}} = \underbrace{-k_S \frac{\partial T^S}{\partial n}\bigg|_{\overline{S_S}}}_{Q^S} \tag{33}$$

To compute $\delta J/\delta b_l$, the continuous adjoint to the bi-fluid CHT problem is derived, following the mathematical development presented in Gkaragkounis et al. (2018), there for single fluid flows. The adjoint mean-flow and turbulence equations as well as the Hamilton-Jacobi equation over the fluid regions $\Omega_F^{(\gamma)}$ are those already presented in Sect. 3.3 (Eqs. 22), stripped from the source terms of TopO. The adjoint energy equations over $\Omega_F^{(\gamma)}$ and Ω_S are

$$R^{T_\alpha^{F^{(\gamma)}}} = -\rho^{(\gamma)} c_p^{(\gamma)} \frac{\partial (v_j^{(\gamma)} T_\alpha^{F^{(\gamma)}})}{\partial x_j} - \frac{\partial}{\partial x_j}\left(k_{eff}^{(\gamma)} \frac{\partial T_\alpha^{F^{(\gamma)}}}{\partial x_j}\right) = 0 \tag{34}$$

$$R^{T_a^S} = -\frac{\partial}{\partial x_j}\left(k_S \frac{\partial T_\alpha^S}{\partial x_j}\right) = 0 \tag{35}$$

In the interest of space, the adjoint boundary conditions for the mean-flow and turbulence equations are not presented here (these can be found in Gkaragkounis et al. (2018)). Only the FSI conditions for T_a are given,

$$\begin{aligned} \left(-T_\alpha^{F^{(\gamma)}} + \frac{\partial J}{\partial Q^{F^{(\gamma)}}}\right)\bigg|_{\overline{S_F}^{(\gamma)}} &= \left(-T_\alpha^S + \frac{\partial J}{\partial Q^S}\right)\bigg|_{\overline{S_S}} \\ \left(k_{eff}^{(\gamma)} \frac{\partial T_\alpha^{F^{(\gamma)}}}{\partial n} + \frac{\partial J}{\partial T^{F^{(\gamma)}}}\right)\bigg|_{\overline{S_F}^{(\gamma)}} &= -\left(k_S \frac{\partial T_a^S}{\partial n} + \frac{\partial J}{\partial T^S}\right)\bigg|_{\overline{S_S}} \end{aligned} \tag{36}$$

The sensitivity derivatives of J w.r.t. b_l are computed through

$$\begin{aligned} \frac{\delta J}{\delta b_l} = \sum_{\gamma=1,2} \int_{\Omega_F^{(\gamma)}} \Bigg[& -u_j^{(\gamma)} \frac{\partial p^{(\gamma)}}{\partial x_k} + q^{(\gamma)} \frac{\partial v_j^{(\gamma)}}{\partial x_k} - v_j^{(\gamma)} u_i^{(\gamma)} \frac{\partial v_i^{(\gamma)}}{\partial x_k} - \tau_{ij}^{\alpha^{(\gamma)}} \frac{\partial v_i^{(\gamma)}}{\partial x_k} \\ & + u_i^{(\gamma)} \frac{\partial \tau_{ij}^{(\gamma)}}{\partial x_k} - \rho^{(\gamma)} c_p^{(\gamma)} v_j^{(\gamma)} T_\alpha^{F^{(\gamma)}} \frac{\partial T^{F^{(\gamma)}}}{\partial x_k} - k_{eff}^{(\gamma)} \frac{\partial T^{F^{(\gamma)}}}{\partial x_k} \frac{\partial T_\alpha^{F^{(\gamma)}}}{\partial x_j} \\ & + T_\alpha^{F^{(\gamma)}} \frac{\partial}{\partial x_k}\left(k_{eff}^{(\gamma)} \frac{\partial T^{F^{(\gamma)}}}{\partial x_j}\right) + W_{SA,jk} \Bigg] \frac{\partial}{\partial x_j}\left(\frac{\delta x_k}{\delta b_l}\right) d\Omega_F^{(\gamma)} \\ & + \int_{\Omega_S} \left[-k_S \frac{\partial T_\alpha^S}{\partial x_j} \frac{\partial T^S}{\partial x_k} + T_\alpha^S \frac{\partial}{\partial x_k}\left(k_S \frac{\partial T^S}{\partial x_j}\right) \right] \frac{\partial}{\partial x_j}\left(\frac{\delta x_k}{\delta b_l}\right) d\Omega_S \\ & + \sum_{\gamma=1,2} \int_{S_{F,W} \cup \overline{S_F}^{(\gamma)}} k_{eff}^{(\gamma)} T_\alpha^{F^{(\gamma)}} \frac{\partial T^{F^{(\gamma)}}}{\partial x_j} \frac{\delta n_j}{\delta b_l} dS \\ & + \int_{S_{S,W} \cup \overline{S_S}} k_S T_\alpha^S \frac{\partial T^S}{\partial x_j} \frac{\delta n_j}{\delta b_l} dS \end{aligned} \tag{37}$$

where $W_{SA,jk}$ are the terms arising from the differentiation of the Spalart–Allmaras model PDE, Papoutsis-Kiachagias et al. (2019).

The ShpO problem, given in its general form by Eq. 1, is setup with Eq. 4 as the objective function and g_M (Eq. 5) as its only constraint. After computing $\frac{\delta J}{\delta b_l}$ and $\frac{\delta g_M}{\delta b_l}$ using Eq. 37, the design variables are updated by, Nocedal and Wright (2001)

$$\Delta b_l = -\eta_{CP}\left(\delta_{lm} - \frac{\frac{\delta g_M}{\delta b_l}\frac{\delta g_M}{\delta b_m}}{\left(\frac{\delta g_M}{\delta b_k}\right)^2}\right)\frac{\delta J}{\delta b_m} - \frac{\frac{\delta g_M}{\delta b_l}}{\left(\frac{\delta g_M}{\delta b_k}\right)^2} g_M \tag{38}$$

where η_{CP} corresponds to a user defined steepest-descent length and δ_{lm} is a Kronecker delta.

6 Optimization of a Bi-fluid heat exchanger

6.1 Geometrical data and flow conditions

The design of a compact counter-flow heat exchanger with an orthogonal $22 \times 22 \times 20mm^3$ design domain (Fig. 2) is presented. The hot fluid ($\gamma = 2$) enters the heat exchanger from eight ($5mm \times 5mm$) square inlets and exits from a single ($10mm \times 10mm$) square outlet; the exact opposite

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configuration is valid for the cold fluid ($\gamma = 1$). The outlets from the cold stream and the inlets to the hot stream are arranged in a checkerboard-like pattern, separated by $0.5mm$ wide solid gaps. Inlets and outlets of both fluids are extended by straight ducts of $10mm$ length to facilitate the convergence of the flow equations.

The cold fluid stream corresponds to water and the hot fluid to air. The thermophysical properties and inlet flow conditions of the two fluids are given in Table 1; the solid material is aluminum ($k_s = 136\frac{W}{m \cdot K}$).

6.2 TopO design

The study starts by using the bi-fluid TopO algorithm to create paths/manifolds connecting inlets with outlets for the two fluids by maximizing heat exchange between them and minimizing total pressure losses. In specific, the optimization is setup with a pair of equal weights, i.e., $w_E = w_P = 0.5$, for the objective functions in Eq. 4, by also imposing the constraints mentioned in Sect. 2. The Darcy-like TopO, used to initialize the α field, aims to minimize Eq. 4 by replacing each $J_P^{(\gamma)}$ with $J_{PL}^{(\gamma)}$ (Eq. 29). For both TopO runs, $V_{target}^{(1)} = V_{target}^{(2)} = 0.35$ are used in the volume constraints of Eq. 6. The TopO grid comprises $1.1M$ hexahedral cells. The impermeability parameter β_{max} is computed, separately for each fluid, based on its kinematic viscosity, using Eq. 8; so, $\beta_{max}^{(1)} = 4000$ and $\beta_{max}^{(2)} = 60000$. To ensure that the two fluid streams remain separated with at least 2 cells, the erosion filter radius r_e (see Sect. 3.2) is set equal to $3.74 \cdot 10^{-4}m$.

6.2.1 TopO initialization

First, the Darcy-like flow model of Sect. 3.4 and its adjoint are used to initialize the design variables (α field), at low-cost. The design space, colored in gray in Fig. 2, starts with solid material ($\alpha = 0.5$) everywhere. The b_H parameter in the second projection step is set to $b_H = 7$ for the first few optimization cycles and gradually increases to $b_H = 28$. Such a treatment evades premature convergence to poor local minima.

The wall clock time for the Darcy-like TopO was $\sim$ 1 hour on 4 AMD EPYC 7452 32-core CPUs (128 cores) for 100 optimization cycles. The areas occupied by the two fluids, identified as areas with $\xi^{(1)} > 0.9$ and $\xi^{(2)} > 0.9$, respectively, for six instants of the optimization, are presented in Fig. 3. The evolution of the objective functions and constraints during the Darcy-TopO are plotted in Fig. 4. $J_{PL}^{(\gamma)}$ take on high values during the first few cycles since the domain is initially blocked by solid material but they immediately start dropping since part of the domain gradually becomes fluidized; the latter is also reflected by the increase in the values of the volume constraint functions. Due to the fact that Ω is initially completely solid and has thus very high power dissipation, the final design achieves a reduction by almost four orders of magnitude in $J_{PL}^{(\gamma)}(\gamma = 1, 2)$, while respecting both volume constraints. Regarding the thermal performance of the heat exchanger, $J_E^{(2)}$ starts with low values in the first optimization cycles (Fig. 4), indicating that high heat transfer takes place; this is, of course, due to the unrealistic all-solid initialization. After flow paths have been carved for both fluids, its value remains almost constant throughout the next cycles. Flow streamlines colored by the velocity and temperature magnitudes for each fluid are shown in Fig. 5 in the so-computed geometry.

The normalized volume flowrates Q_{O_k}/Q_I, Eq. 5, at the $k \in [1, 8]$ outlets of the cold fluid stream are plotted for various optimization cycles in Fig. 6. In the first cycle, the cold stream outlet flowrates appear as equi-distributed, but this has little impact due to the unrealistic initialization. In cycles 10 and 20, only some of the cold stream outlets are connected to the inlet, and the flow exits primarily through them; this is rectified in the converged solution, in which an almost perfect (at least according to the low-fidelity TopO evaluation model) flowrate balance among all outlets is

Table 1 Thermophysical properties and flow conditions of the two fluids

Property	Cold (water; $\gamma = 1$)	Hot (air; $\gamma = 2$)
Thermal Conductivity $k\left[W \cdot m^{-1} \cdot K^{-1}\right]$	0.598	0.0228
Density $\rho\left[kg \cdot m^{-3}\right]$	998.3	1.25
Specific Heat Capacity $c_p\left[J \cdot kg^{-1} \cdot K^{-1}\right]$	4184	1006
Kinematic viscosity $\nu\left[m^2 \cdot s^{-1}\right]$	$1 \cdot 10^{-6}$	$1.5 \cdot 10^{-5}$
Inlet velocity $\lvert v_{inlet} \rvert\left[m \cdot s^{-1}\right]$	0.256	0.96
Inlet width $H[m]$	0.01	0.005
Number of inlets/outlets $[-]$	1/8	8/1
Mass flowrate $\dot{m}\left[kg \cdot s^{-1}\right]$	$2.556 \cdot 10^{-2}$	$2.88 \cdot 10^{-5}$
Inlet temperature $T\left[^\circ C\right]$	20	60

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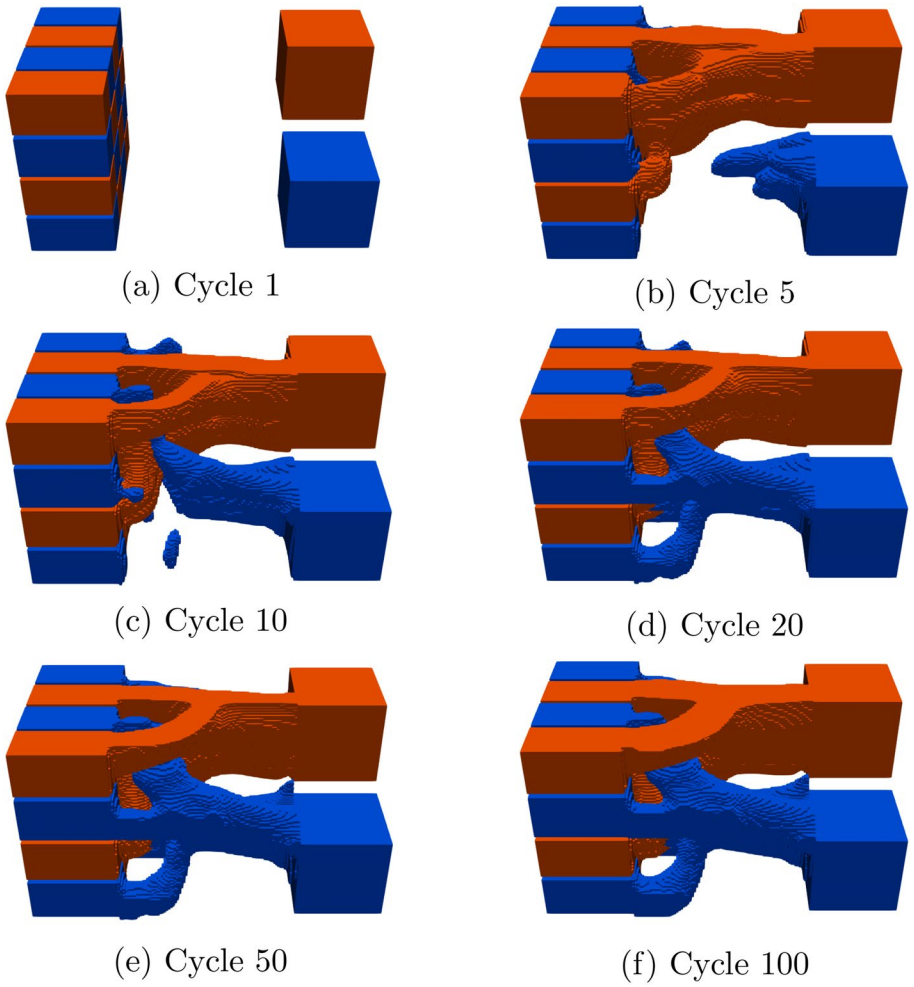
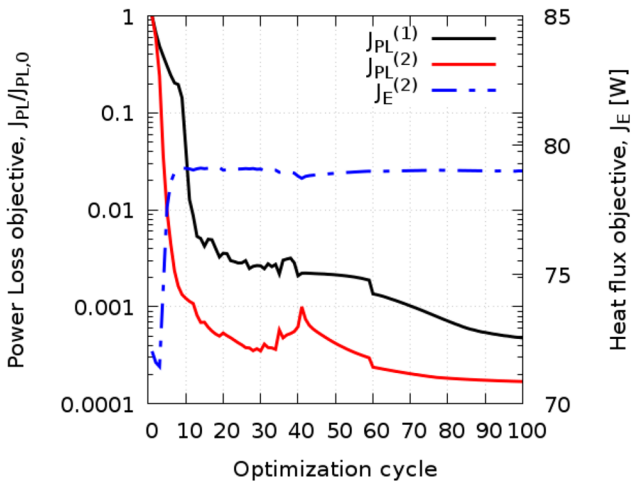


Fig. 3 Evolution of the Darcy-like TopO: Areas occupied by the hot stream (red) entering Ω from the left and the cold stream (blue) entering from the right



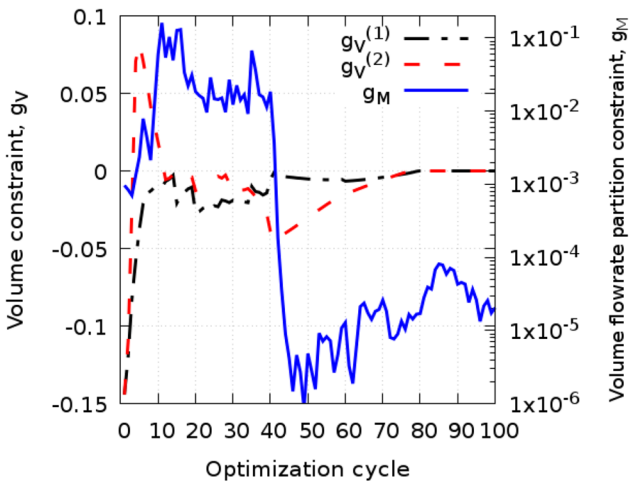


Fig. 4 Darcy-like TopO: Left: evolution of the power dissipation objectives $J_{PL}^{(\gamma)}$, $\gamma = 1, 2$ (Eq. 29; the vertical axis is in logarithmic scale) for both fluids and the energy flux objective $J_E^{(2)}$ (Eq. 2) for the hot fluid. Right: evolution of the volume constraints $g_V^{(\gamma)}$, $\gamma = 1, 2$ (Eq. 6) and constraint g_M associated with the equi-distribution of outgoing flowrates (Eq. 5; in logarithmic scale)

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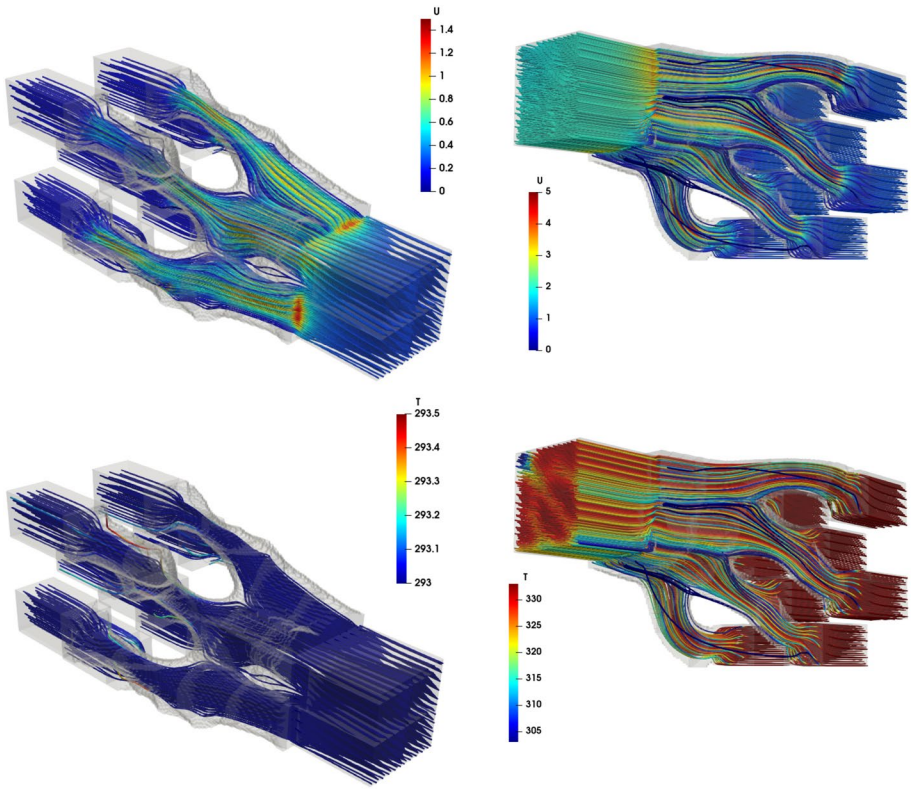


Fig. 5 Flow streamlines of the cold (left) and hot (right) fluid for the outcome of the Darcy-like TopO. Velocity magnitudes (top) and temperatures (bottom)

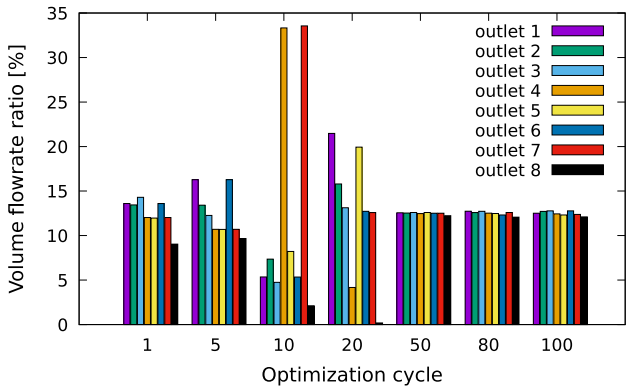


Fig. 6 Darcy-like TopO: Ratios of the 8 outgoing volume flowrates to the incoming flowrate of the cold stream for various optimization cycles. The ideal value for all ratios is $100/8 = 12.5\%$

achieved. This solution is used to initialize the RANS-based CHT TopO that follows.

6.2.2 RANS-based CHT TopO

The Darcy-like TopO optimized solution is firstly re-evaluated using the RANS-based TopO solver which, according to Fig. 8, computes significantly different velocity magnitude and temperature fields compared to the Darcy flow solution. Regarding the velocity, these differences can be attributed to the inviscid Darcy flow model. From Fig. 8, it can be deduced that the Darcy-like flow solution overestimates the hot fluid temperature. Additionally, the re-evaluation shows that the cold fluid is no longer equi-distributed at its outlets (see Fig. 7). Based on the above, it is obvious why the Darcy-like TopO is used only for the purpose of initializing the α field and RANS-based TopO should follow.

The RANS-based TopO terminated after 50 cycles at the cost of $\sim$ 1184.4 CPU hours (wall clock time $\sim$ 9.3 hours on the same 128 cores). CPU cost is by one order of magnitude higher than that of the Darcy-like TopO. The termination of the RANS-based TopO with relaxed convergence criteria for the objective and constraint functions (Fig. 9), was decided since the TopO optimized solution will be followed by ShpO.

The evolutions of the $J_P^{(\gamma)}$ objective values and the g_M constraint are plotted in Fig. 9. In the optimized solution, both $J_P^{(\gamma)}$ objectives are reduced by more than 70% compared to the outcome of the Darcy-like TopO. This is, however, at the expense of a slightly worse cooling performance as indicated by the increase in $J_E^{(2)}$ in Fig. 9. According to Figs. 7 and 9, a great imbalance in the volume flowrates at the cold stream outlets occurs in the initialization of the RANS-based TopO; this is, however, mitigated within a few optimization cycles, leading to a practically perfect balance. The $g_V^{(\gamma)}$ constraints

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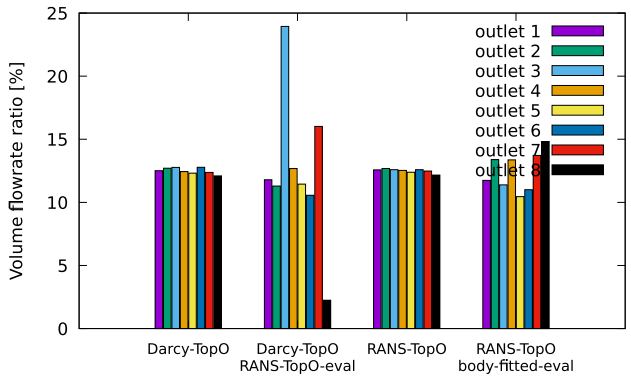


Fig. 7 Ratios of the 8 outgoing volume flowrates to the incoming flowrate of the cold stream computed at four phases of the design process. Columns from left to right correspond to: (a) the outcome of the Darcy-like TopO initialization, evaluated on its own solver, (b) same topology, evaluated on the RANS-based TopO solver, (c) the outcome of the RANS-based TopO, (d) the outcome of the RANS-TopO re-evaluated on a RANS solver using a body-fitted mesh

are also met throughout the optimization. Streamlines for both working fluids are presented in Fig. 12, colored based on temperature and velocity magnitudes.

The $\xi^{(1)} = 0.9$ and $\xi^{(2)} = 0.9$ iso-surfaces, corresponding to the computed side-walls of the two fluid channels, are plotted in Fig. 10 for the starting solution obtained from the Darcy-like TopO and the outcome of the RANS-based TopO. In the starting geometry, the entrance of the cold fluid in the design domain and the exit of the hot fluid from it are partially blocked by solid material. This is rectified in the final solution, which explains the reduction in $J_P^{(1)}$ and $J_P^{(2)}$. In Fig. 11, the $1 - \xi^{(1)} - \xi^{(2)}$ indicator field is plotted at three cross-sections along the length of the heat exchanger, showcasing that the method results in well-defined fluid channels, separated by solid material of adequate thickness.

6.3 Body-fitted grid evaluation of the TopO solution

The TtoST process, described in Sect. 4, helps generating body-fitted grids based on the RANS-based TopO solution. The threshold values for the $\xi^{(1)}$ and $\xi^{(2)}$ fields used in the first step of the TtoST process were set to $\eta^{(1)} = \eta^{(2)} = 0.9$. After creating the subset grids for each fluid and solid, a radius $r_m = 3 \cdot 10^{-4} m$ (Eq. 30) was used to smooth the FSI boundaries, computed as the average edge length of the non-smooth FSIs. Then, the smoothed boundaries were triangulated and stored in STL format; they are shown in Fig. 13.

The body-fitted grids for the two fluids, generated according to the extracted FSI, comprise $\sim 2.1M$ cells each, and are stretched close to the FSI boundaries; the solid grid has $\sim 3M$ cells. The flow and heat transfer problems are solved on these grids and, in Fig. 15, temperature and velocity fields are plotted as a function of the y-coordinate along a line positioned at $x = 8.25$ mm, $z = -10$ mm. This line passes through the center of the design space and intersects both fluid paths. Small deviations in the computed velocity fields between the two (TopO and body-fitted) RANS evaluation tools can be seen. Despite the inability of TopO to impose exact no-slip boundary conditions over the FSI wall, the TopO flow solver is able to decently capture the flow boundary layer. The temperature field computed by means of the TopO evaluation tool takes on smaller values in the hot fluid stream, which translates to more heat transferred from the cooled fluid to the coolant; this can be confirmed in Table 2, according to which TopO computes a higher value for $J_E^{(1)}$ (Fig. 14). Still, a quite good agreement is achieved considering the fact that, in TopO, thermophysical properties are interpolated close to the FSI (Eq. 11) where $\xi^{(1)}$ and $\xi^{(2)}$ take on intermediate values, instead of imposing the strict FSI conditions (Eqs. 33). To investigate the effectiveness of the material interpolation scheme used in TopO (Eq. 11), the thermal conductivity k_{eff} is, also, plotted in Fig. 15.

A comparison between the objective values computed using the RANS-based TopO and a body-fitted RANS solver is shown in Table 2, where some discrepancies occur despite the relatively good agreement in the computed flow fields (Fig. 15). With regard to $J_P^{(\gamma)}$ and $\Delta p_t^{(\gamma)}$ ($\gamma = 1, 2$), the TopO inaccuracy w.r.t. the ground truth (i.e., evaluation on a body-fitted grid) is of the order of 10%. The volumes occupied by each of the fluids are, also, computed. The TtoST process resulted in fluid regions which more or less preserve the volume of both fluids during the last TopO iteration. Additionally, the evaluation of the TopO solution on a body-fitted grid indicates that the constraint of distributing the cold fluid uniformly to its many outlets is, in fact, not met (Fig. 7).

6.4 CHT ShpO design

The fluid and solid regions extracted by TtoST are subjected to a ShpO to be further refined. The new design variables are the coordinates of the control points of a single volumetric B-Splines morphing lattice, the boundaries of which coincide with the boundaries of the rectilinear part of Ω (see Fig. 16), parameterizing all grid nodes redising within it. The control lattice consists of $12 \times 12 \times 11$ control points (along the x, y and z directions, respectively), with a degree of 3 for the B-Spline basis functions. The outer control points remain fixed to ensure that the design space boundaries do not change. The optimization is setup with the same weights as the RANS-based TopO, while imposing the constraint pertaining to the equal distribution of the coolant at its outlets. The minimum solid thickness separating the two fluid streams is not strictly preserved during the ShpO. A way to ensure minimum solid length-scale would be by including an additional geometric constraint. However, in

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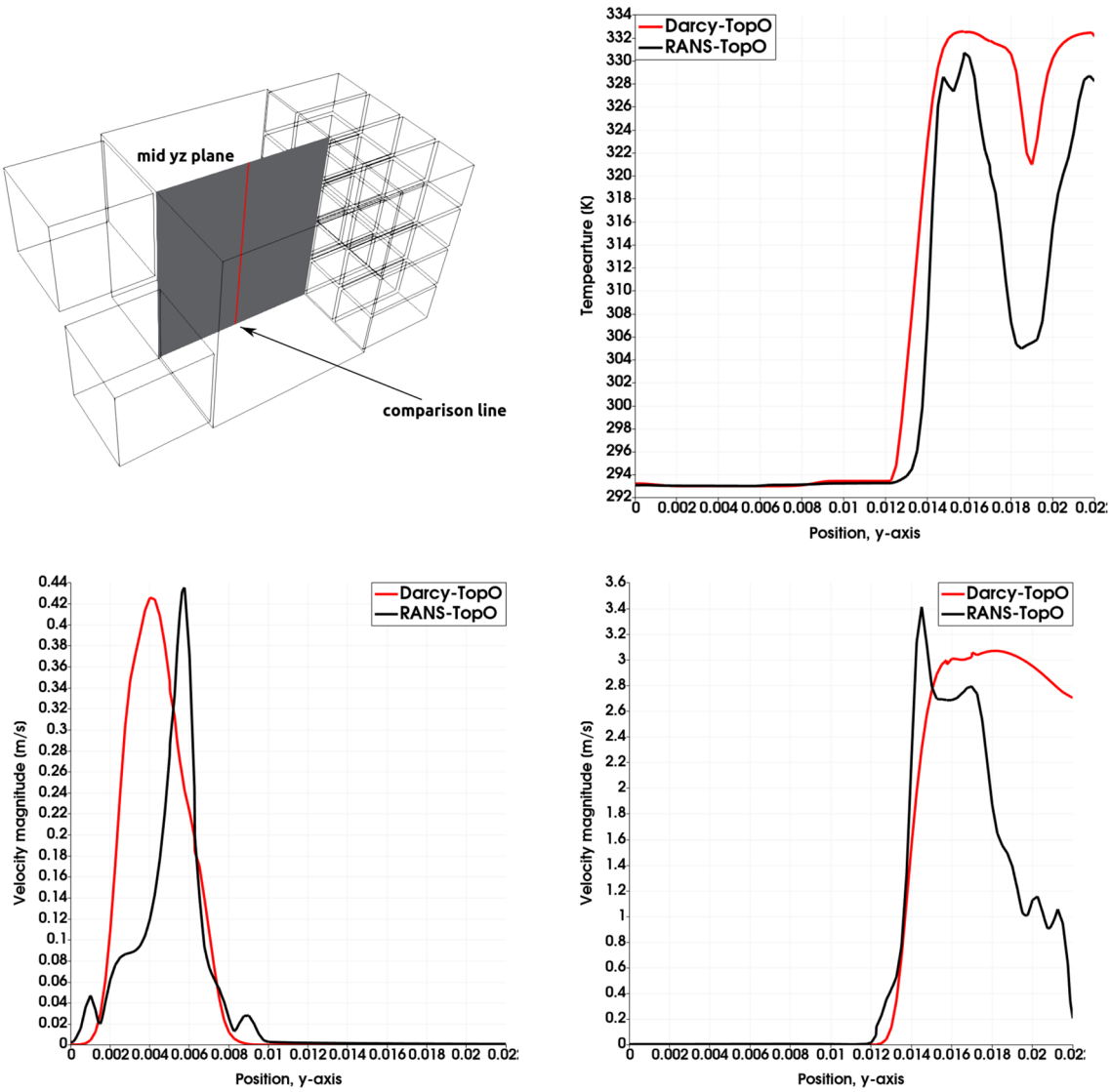


Fig. 8 Comparison of the velocity and temperature magnitude distributions computed by the Darcy-like and the RANS-based TopO solvers along the line shown on the top left

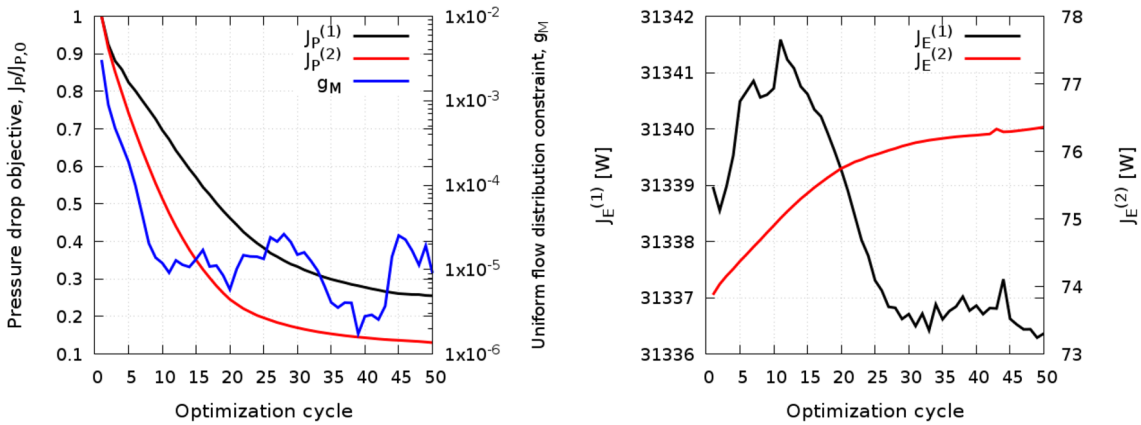


Fig. 9 RANS-based TopO: Left: evolution of the total pressure drop objective functions (J_P) and the uniform flowrate distribution constraint (g_M), the latter plotted in a logarithmic scale. Right: evolution of the heat flux (J_E) defined at the outlets from the cold and hot stream

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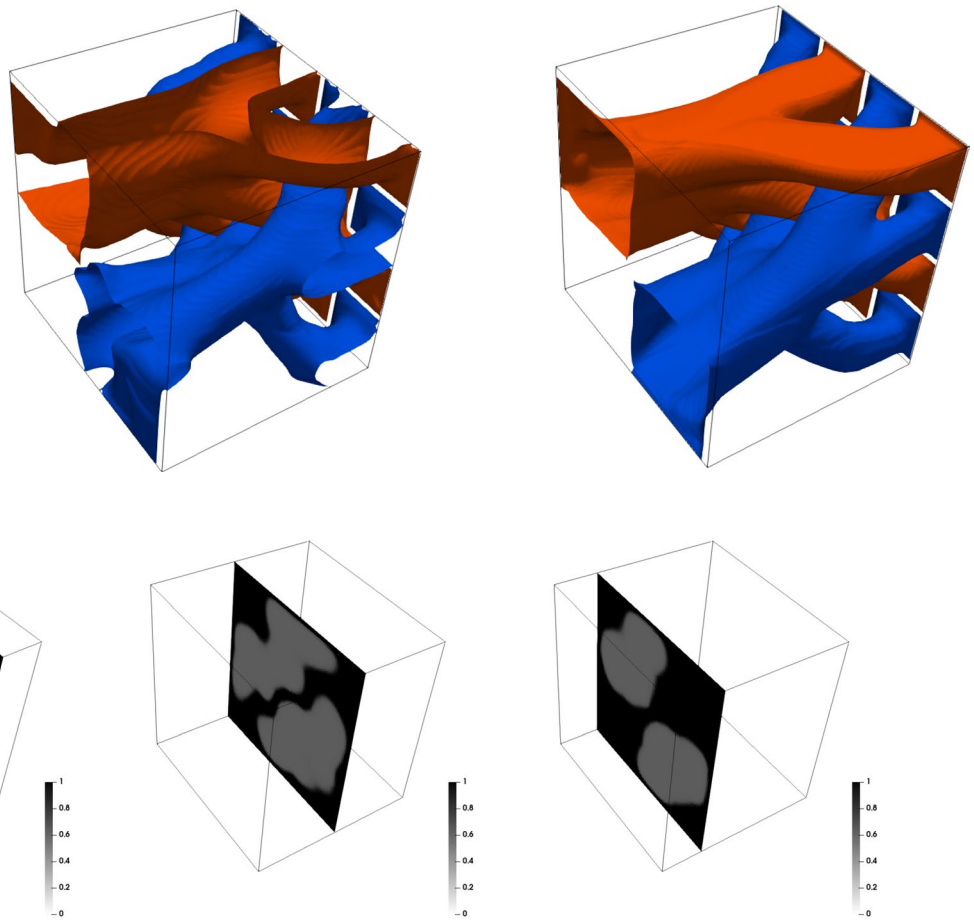


Fig. 10 $\xi^{(1)} = 0.9$ and $\xi^{(2)} = 0.9$ iso-contours indicating the FSI boundaries. Left: the FSI of the outcome of the Darcy-like TopO, Right: the FSI of the outcome of the RANS-based TopO

Fig. 11 Outcome of the RANS-based TopO: the $1 - \xi^{(1)} - \xi^{(2)}$ field, indicating the presence or absence of solid material, plotted over three transversal cuts along the length of the heat exchanger

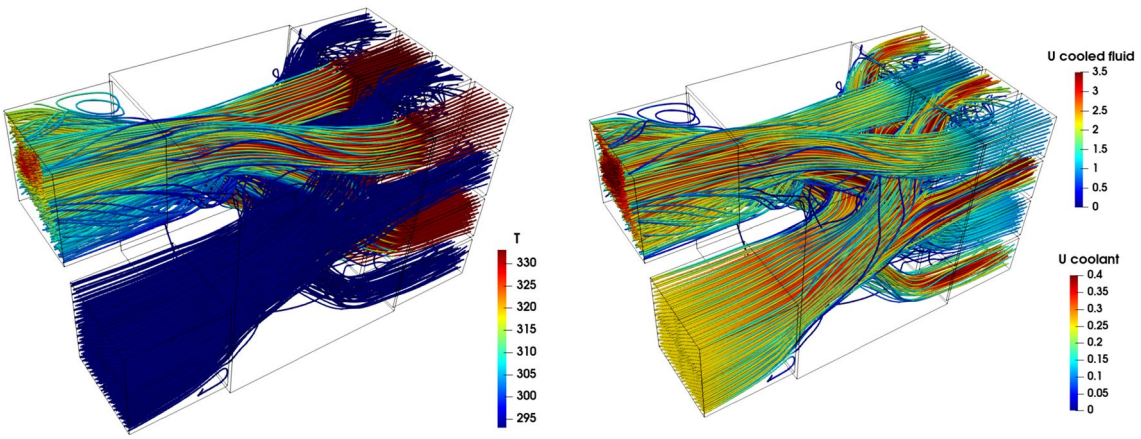


Fig. 12 RANS-based TopO: Flow streamlines, colored by the temperature (left) and velocity magnitude (right), for the outcome of RANS-based TopO

such an application, this proved to be an unnecessary burden since, the volumetric B-Splines parameterization could by itself retain the desired thickness (see quantified deviations in a subsequent paragraph).

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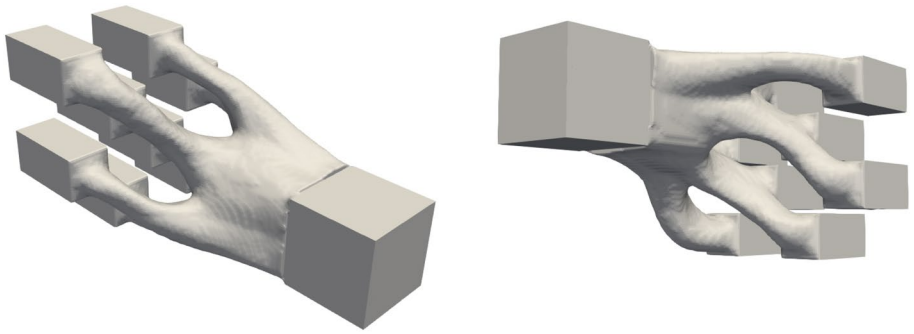
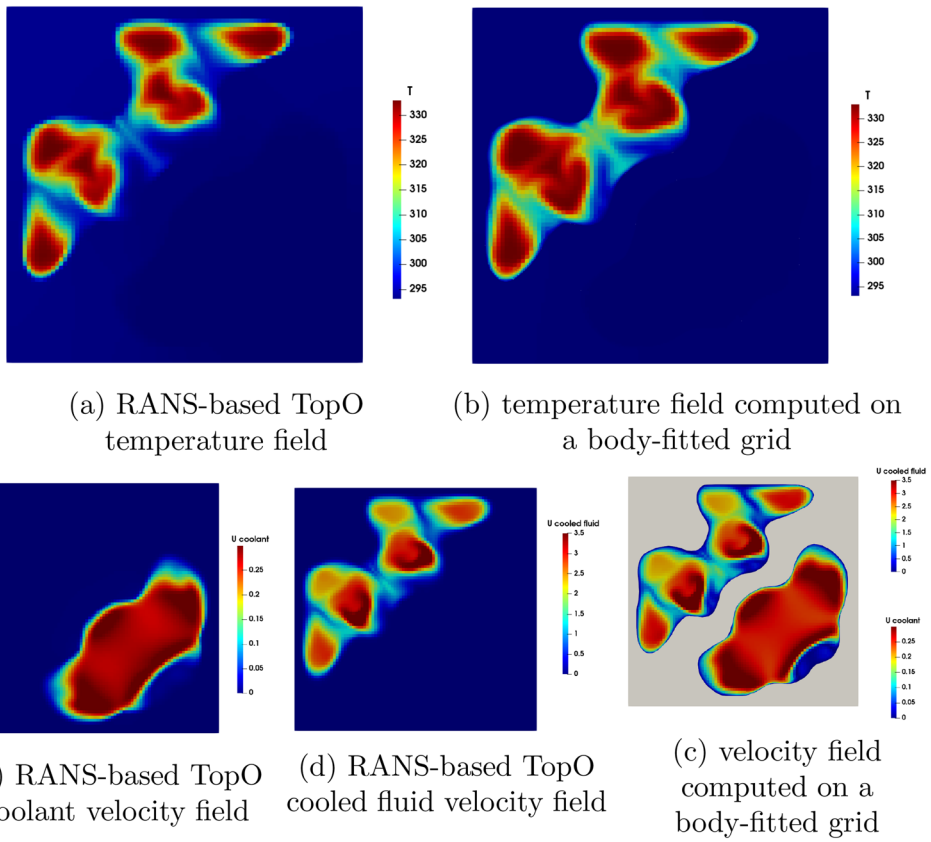


Fig. 13 TtoST: The nodal representation of the fluid region boundaries computed for the outcome of the RANS-based TopO



(a) RANS-based TopO temperature field

(b) temperature field computed on a body-fitted grid

(c) RANS-based TopO coolant velocity field

(d) RANS-based TopO cooled fluid velocity field

(c) velocity field computed on a body-fitted grid

Fig. 14 Temperature and velocity fields plotted at the mid xy-plane ($z = -10mm$) for the outcome of the RANS-based TopO. Left: evaluation on the RANS-TopO solver. Right: evaluation on the RANS solver using body-fitted grids

A total of 15 optimization cycles were carried out. ShpO was performed on the same hardware as TopO, and the computational cost associated with it was $\sim$ 1890.6 CPU hours (wall clock time $\sim$ 15 hours on 128 cores). The convergence of the non-normalized $J_P^{(\gamma)}$ objective functions throughout the whole optimization (both RANS-based TopO and ShpO) is plotted in Fig. 17; in ShpO, their values are reduced (compared to the geometry generated by TtoST) by 35% for $\gamma = 1$ and 25% for $\gamma = 2$. In Fig. 18, the displacements of all FSI nodes of the two fluids are projected onto $\mathbf{n}|_{\overline{S_F}^{(\gamma)}}$ and plotted on the optimized geometry. Most areas have been pushed outwards from the corresponding fluid domain, decelerating the flow and reducing the total pressure drop. Note that, this reduction in power consumption is achieved without damaging the cooling performance of the heat exchanger, since the energy flux of the hot air at its outlet ($J_E^{(2)}$) remains practically the same during ShpO (see Fig. 17). This, practically, translates to the same amount of heat exchanged through the FSI boundaries. This conclusion can, also, be deduced from Fig. 19, where it can be seen that, overall, the heat flux difference between the TtoST and ShpO optimized geometries, evaluated at the hot fluid FSI, is quite balanced. Finally, the solid material thickness that separates the two fluids is computed for the starting geometry of the ShpO run and the optimized one. For the starting heat exchanger geometry its value reads $7.6 \cdot 10^{-4} m$ whereas for the optimized one $4.94 \cdot 10^{-4} m$. The latter is quite close to the minimum solid

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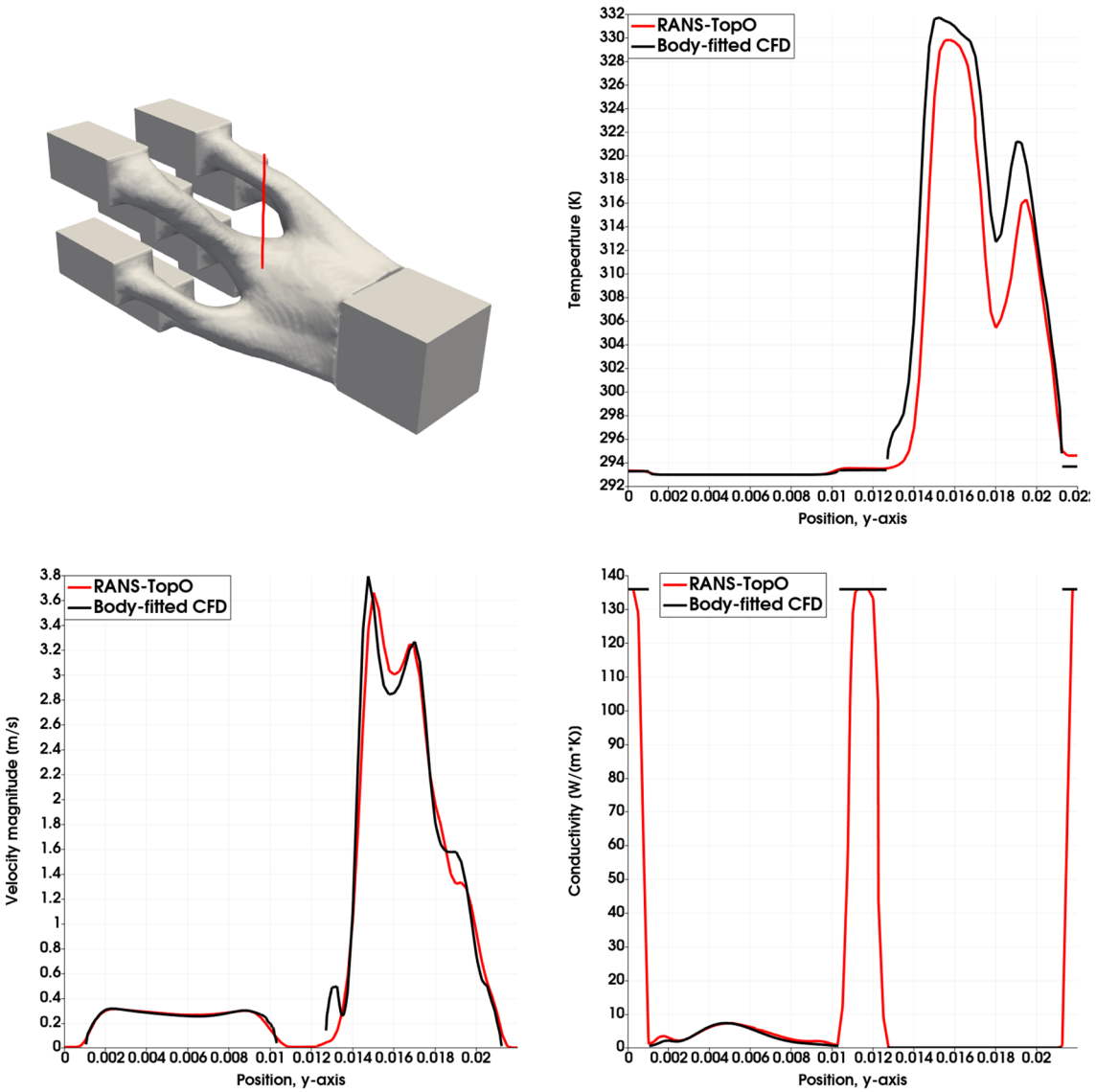


Fig. 15 Outcome of the RANS-based TopO: Temperature, velocity and thermal conductivity evaluated with the RANS-based TopO software and with body-fitted grids as a function of the y-coordinate at $x = 8.25mm, z = -10mm$

Table 2 Outcome of RANS-based TopO: Objective values and related quantities evaluated on the RANS-based TopO and the RANS (body-fitted grids) solvers

Quantity of Interest	TopO evaluation		Body-fitted grid evaluation	
	Cold fluid (water; $\gamma = 1$)	Hot fluid (air; $\gamma = 2$)	Cold fluid (water; $\gamma = 1$)	Hot fluid (air; $\gamma = 2$)
$J_P^{(\gamma)}$ $[10^{-4}m^5 \cdot s^{-3}]$(Eq. 3)	0.0160098	6.348357	0.0166186	5.746771
$\Delta p_t^{(\gamma)}$ [Pa]	83.432	5.793	90.182	5.201
$J_E^{(\gamma)}$ $[W]$ (Eq. 2)	31341.18	76.42	31298.04	77.74
$\Delta T^{(\gamma)}$ $[°C]$	0.49	20.75	0.64	14.45
$V^{(\gamma)}$ $[10^{-6}m^3]$	5.4868	5.4834	5.1813	5.1741

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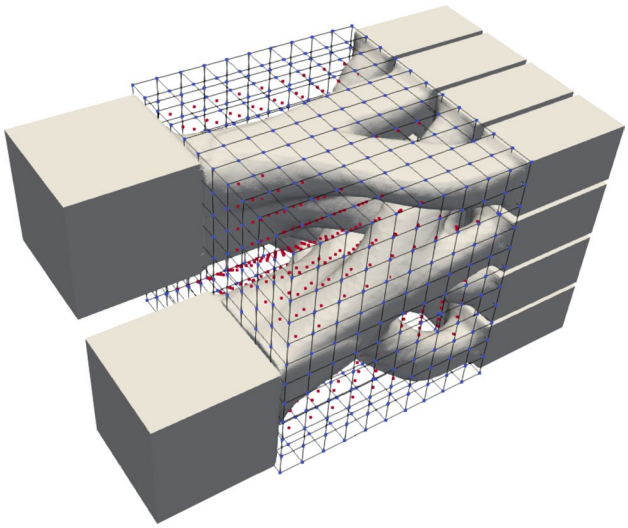


Fig. 16 Volumetric B-Splines-based control box used to parameterize the two fluids' and solid regions, during ShpO

length-scale of $5 \cdot 10^{-4}m$, which was imposed during both the Darcy- and the RANS-TopO.

Despite having obtained a good heat exchanger design that performs well in terms of both power consumption and cooling performance, an additional ShpO was performed starting, also, from the same outcome of TtoST. In order to attain an improvement in the thermal performance, compared to the outcome of TtoST, the new ShpO seeks to minimize solely $J_E^{(2)}$, by using $w_E = 1$ and $w_P = 0$ in Eq. 4. This run will be referred to as ShpO-P2, to distinguish it from the previously conducted ShpO. After 5 optimization cycles (see Fig. 20 for the evolution of objective and constraint quantities), ShpO-P2 reduces $J_E^{(2)}$ by $\sim 0.7W$ ($\sim 1\%$), at the expense of increasing $J_P^{(2)}$ by more than 3 times compared to the geometry that it started from. An unexpected result is that the total pressure drop in the cold stream is reduced by 20%, even though $J_P^{(1)}$ was not a target in this optimization. To better understand the heat transfer mechanism, the nodal displacements of the cold and hot stream FSI are projected onto $\mathbf{n}|_{\overline{S_F}^{(\gamma)}}$ and plotted in Fig. 21. For the cooled fluid, a great part of the FSI has been pushed inwards to accelerate the flow, enhance turbulence and, thus, heat transfer. On the other hand, the FSI of the coolant is pushed outwards (Fig. 22), at least for most of its part, explaining the drop in $J_P^{(1)}$. For the outcome of ShpO-P2, the minimum solid material thickness that separates the two fluids is computed equal to $8.2 \cdot 10^{-4}m$, which is higher than that of the starting ShpO-P2 geometry. The two ShpO runs highlight the contradictory nature of total pressure losses and heat transfer. A number of optimized geometries can be designed by giving different values to w_E and w_P in Eq. 4, leading eventually to a Pareto front of non-dominated solutions.

7 Summary–conclusions

A self-contained methodology to design bi-fluid heat exchangers via the synergistic use of porosity-based TopO and ShpO was presented. Both are based on continuous adjoint to compute the gradient of objective and constraint functions. The adjoint to the Spalart–Allmaras turbulence model equation and the Hamilton-Jacobi equation

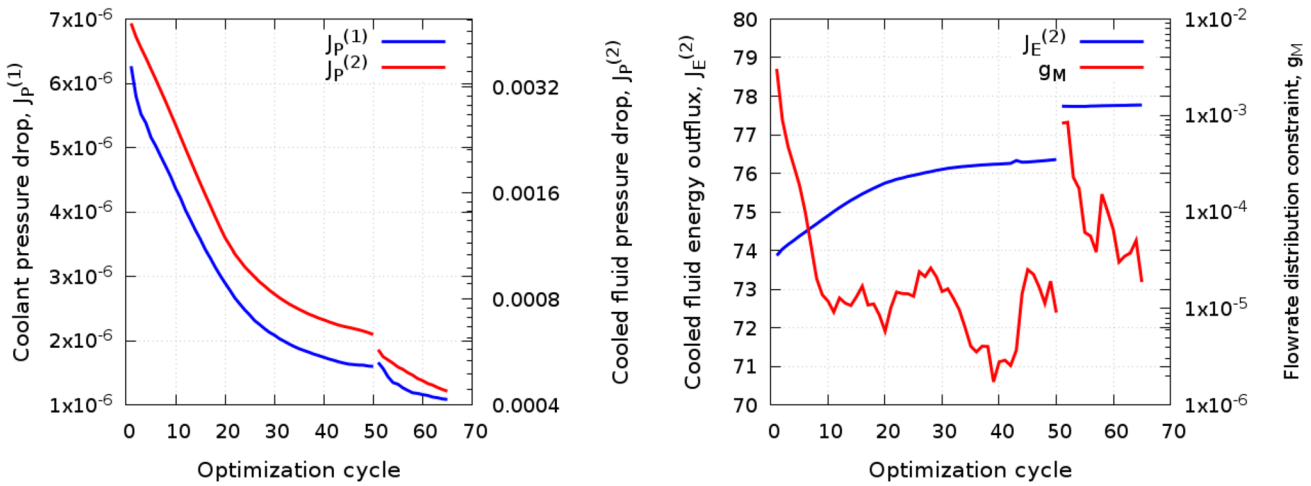


Fig. 17 Successive RANS-TopO (excl. the Darcy-like initialization) and ShpO: Left: evolution of the $J_P^{(\gamma)}$ objective functions values Right: evolution of the $J_E^{(2)}$ objective function (Eq. 2) and g_M constraint (Eq. 5) Around the 50-th cycle, the discontinuity shown indicates that this is the starting point of ShpO

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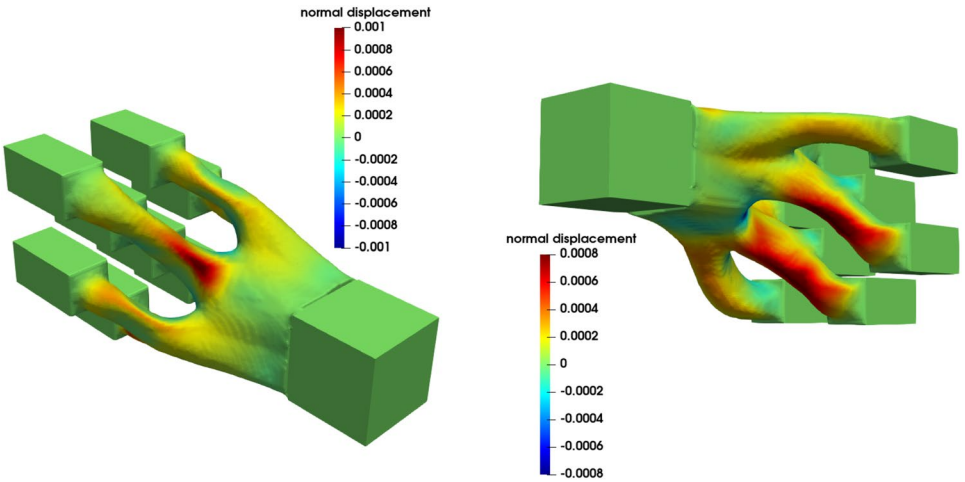


Fig. 18 ShpO optimized solution: FSI nodal displacements of the cold (left) and hot (right) stream projected on $\mathbf{n}|_{\overline{S_F}^{(\gamma)}}$. Red areas have been pushed outwards w.r.t. each of the fluid domains. The opposite holds for blue areas

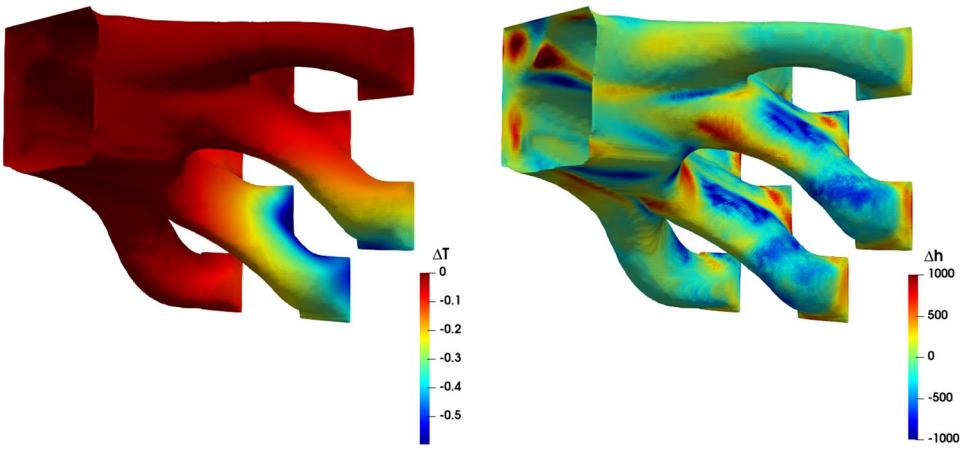


Fig. 19 ShpO optimized solution: Temperature (left) and heat flux (right) differences of the ShpO optimized geometry from the geometry generated by TtoST are plotted on the FSI of the hot fluid. Positive heat flux differences indicate areas where heat transfer from the hot fluid to the coolant has increased

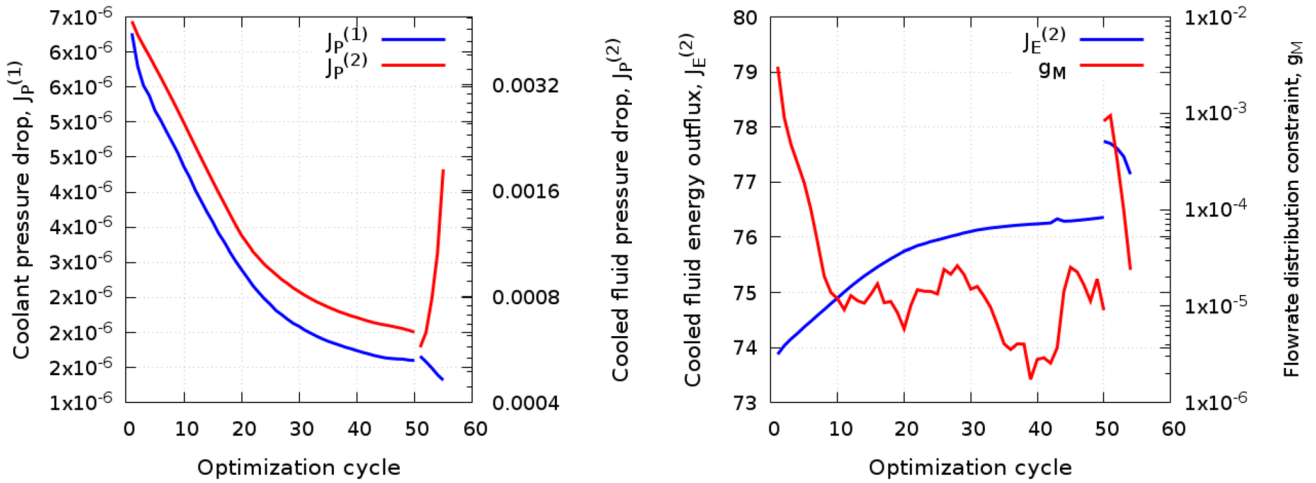


Fig. 20 Successive RANS-TopO and ShpO-P2: Nomenclature as in Fig. 17

that computes distances (even in TopO) are included. The proposed method alleviates the well-known downside of TopO, namely its inability to apply boundary conditions in the strong sense along the sought FSIs, which becomes pronounced in case heat transfer is also considered. To reduce these inaccuracies, a new interpolation scheme for heat conductivity, that becomes effective along the FSI, was proposed for use in TopO runs. To reduce the overall cost

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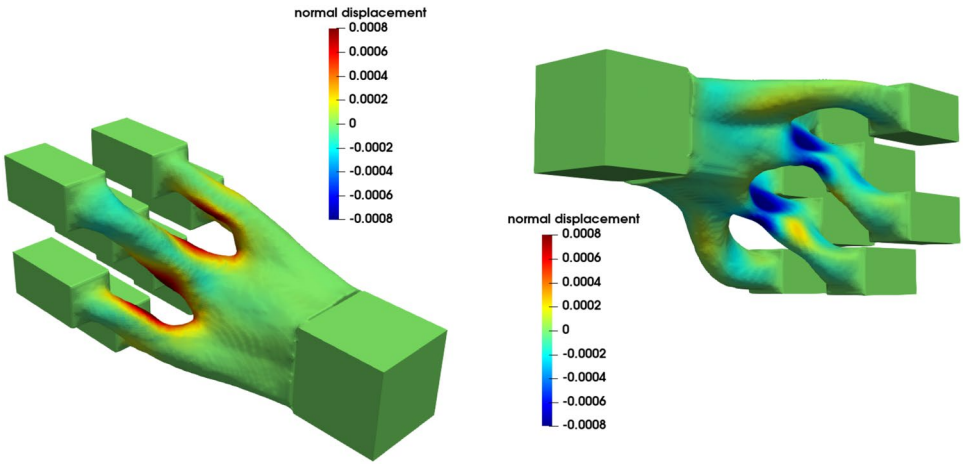


Fig. 21 ShpO-P2 optimized solution: FSI nodal displacements of the cold (left) and hot (right) stream projected on $\mathbf{n}|_{\overline{S_F}^{(\gamma)}}$. Red areas have been pushed outwards w.r.t. each of the fluid domains. The opposite holds for blue areas

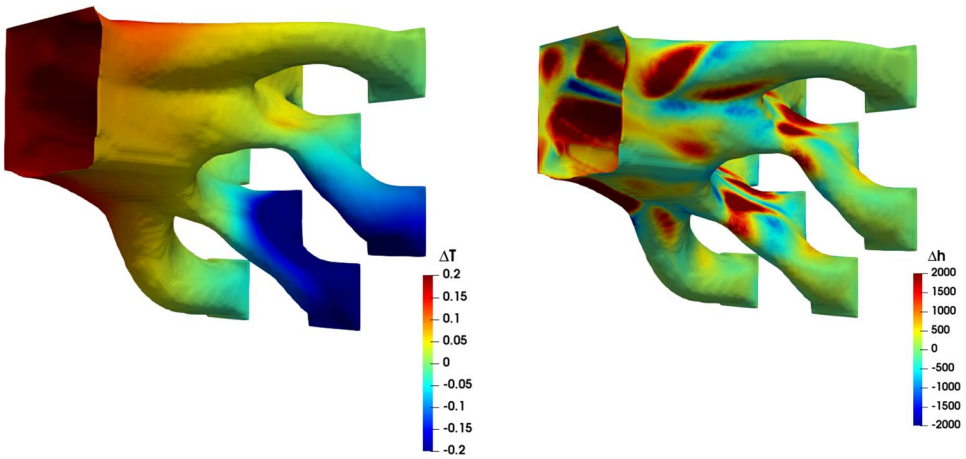


Fig. 22 ShpO-P2 optimized solution: Temperature (left) and heat flux (right) differences of the ShpO-P2 optimized geometry from the geometry generated by TtoST are plotted on the FSI of the hot fluid. Positive heat flux differences indicate areas where heat transfer from the hot fluid to the coolant has increased

of TopO, this includes two parts: (a) a low-cost, low-fidelity TopO based on a simplified Darcy-based flow model and, (b) the high-fidelity RANS-based TopO. An indispensable part of the algorithm is the TtoST, i.e., an intercalary task which exports the FSIs of the optimized solution of the bi-fluid TopO run into a CAD format, which can be used to either manufacture it or generate the body-fitted grids to be used in ShpO. The bi-fluid CHT ShpO was herein performed by using a morphing technique based on volumetric B-Splines; however, any other shape parameterization scheme, including the direct control of the nodal coordinates on the computed FSIs, could be used instead. The proposed methodology was successfully applied to the design of a compact bi-fluid heat exchanger where, starting with a completely solidified design space, optimized manifolds for the two working fluids were obtained.

From the examined case, useful conclusions can be drawn. The flow solution computed by the Darcy-like model deviates significantly from the one computed by the RANS-based TopO, on the same geometry, especially w.r.t. the (im) balance of mass flow rates at the many outlets of the cold fluid. This comment agrees with the findings of Pollini et al. (2020). Nevertheless, its considerably smaller cost (about an order of magnitude) compared to RANS-TopO, attests its usefulness as an initialization tool. Comparing the flow solution of RANS-based TopO and the one obtained from a body-fitted solver on the same geometry gave smaller, but non-negligible, discrepancies, providing a good incentive for refining the designed heat exchanger with ShpO. In specific, total pressure losses differ by $\sim 10\%$ between the two flow solvers, with smaller differences observed in temperature dependent quantities, namely $J_E^{(\gamma)}$ (Eq. 2) and $\Delta T^{(\gamma)}$ ($\gamma = 1, 2$). The ShpOs performed using two different weight combinations for total pressure losses and heat transfer highlighted the contradicting nature of the two widely used performance metrics when designing heat exchangers.

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Appendix A: Summary of the turbulence model functions and constants

Terms (functions and constants) appearing in the Spalart–Allmaras turbulence model are summarized. The eddy viscosity is given by

$$\nu_t = f_{v1}\widetilde{\nu} \tag{A1}$$

where $\widetilde{\nu}$ is the turbulence model variable, computed via the solution of Eq. 7c. The production and dissipation terms in Eq. 7c are given by

$$P(\widetilde{\nu},\Delta) = c_{b1}\widetilde{Y}\,, \quad D(\widetilde{\nu},\Delta) = c_{w1}f_w\frac{\widetilde{\nu}}{\Delta^2} \tag{A2}$$

where $\widetilde{Y}$ is computed through

$$\widetilde{Y} = Y + \frac{f_{v2}\widetilde{\nu}}{\Delta^2\kappa^2}\,, \quad Y = \sqrt{2}\left|\frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right)\right| \tag{A3}$$

Eqs. A2, A3 include the distance Δ from the nearest solid wall, computed by solving Eq. 7d.

The turbulence model functions read

$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}\,, \quad f_{v2} = 1 - \frac{\chi}{1+\chi f_{v1}}\,, \quad \chi = \frac{\widetilde{\nu}}{\nu}$$

$$f_w = g\left[\frac{1+c_{w3}^6}{g^6 + c_{w3}^6}\right]^{\frac{1}{6}}, g = r + c_{w2}\left(r^6 - r\right)\,, \quad r = \frac{\widetilde{\nu}}{\widetilde{Y}\kappa^2\Delta^2}$$

and the constants are $c_{b1} = 0.1355$, $c_{b2} = 0.622$, $\sigma = 2/3$, $\kappa = 0.41$, $c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}$, $c_{w2} = 0.3$, $c_{w3} = 2$, $c_{v1} = 7.1$ and $c_{v2} = 5$.

Appendix B: Effective heat conductivity at the FSI

This section provides the k_{eff} formula at the FSI, when using body-fitted grids. This formula is useful for deriving Eq. 11, which provides an interpolation scheme for k_{eff} used in TopO, for the finite volume faces that lay between cells corresponding to either of the fluids and the solid (Fig. 23 with N corresponding to fluid F, O to solid S and the FSI to their interface f).

The FSI conditions associated with CHT are given by Eq. 33. The discretization of the surface normal gradients included in the second condition, taking the first condition of Eq. 33, also, into account, yields

$$Q^f = -k^F\frac{T^f - T^F}{\mathcal{D}^{Ff}} = k^S\frac{T^f - T^S}{\mathcal{D}^{Sf}} \tag{B1}$$

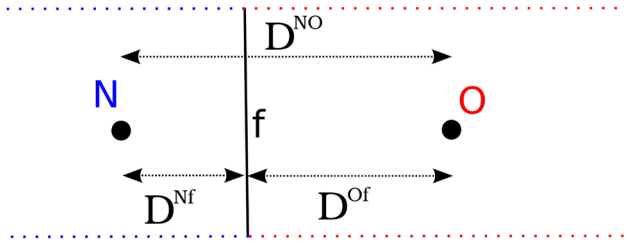


Fig. 23 Schematic of two neighboring cells; their centers are located at N and O, their interface is f; distances between the two cell centers ($\mathcal{D}^{NO}$) and distances between the cell centers and f ($\mathcal{D}^{Nf}$, $\mathcal{D}^{Of}$) are also shown

where Q^f is the heat flux at the FSI, divided by the face area (to be referred to as just heat flux in this appendix). Solving for T^f leads to

$$T^f = \frac{T^F k^F \mathcal{D}^{Sf} + T^S k^S \mathcal{D}^{Ff}}{k^F\mathcal{D}^{Sf} + k^S\mathcal{D}^{Ff}} \tag{B2}$$

Introducing Eq. B2 to either side of Eq. B1 gives an expression for Q^f that depends only on the temperatures of the cells neighboring the FSI, i.e.,

$$Q^f = -\underbrace{\frac{k^S k^F \mathcal{D}^{SF}}{k^F\mathcal{D}^{Sf} + k^S\mathcal{D}^{Ff}}}_{k^f_{eff}}\frac{T^S - T^F}{\mathcal{D}^{SF}} \tag{B3}$$

When discretizing the diffusion term in Eq. 10, the heat flux at any face f is given by

$$Q^f = -k^f_{eff}\frac{T^N - T^O}{\mathcal{D}^{NO}} \tag{B4}$$

If, for instance, N corresponds to the solid side S and O to the fluid side F, then f corresponds to the FSI and k^f_{eff} in Eq. B4 should be equal to that of Eq. B3 in order for TopO to compute the correct heat flux. This provides the motivation for including the last term on the RHS of Eq. 11.

Appendix C: Adjoint field equations and boundary conditions

The field adjoint equations and boundary conditions that emerge from the development of the continuous adjoint method for the bi-fluid CHT TopO problem of Sect. 3.1 are given.

For a general objective function or constaint, J, its derivative w.r.t. the α field of TopO is given by Eq. 20. The R terms, multiplying the derivatives of the flow variables w.r.t. α in the first integral of Eq. 20, are set to zero and give rise to the field adjoint equations which are

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$$R^{q^{(\gamma)}} = -\frac{\partial u_j^{(\gamma)}}{\partial x_j} = 0 \tag{C1a}$$

$$\begin{aligned} R_i^{u^{(\gamma)}} = &-\frac{\partial (v_j^{(\gamma)} u_i^{(\gamma)})}{\partial x_j} + u_j^{(\gamma)} \frac{\partial v_j^{(\gamma)}}{\partial x_i} - \frac{\partial \tau_{ij}^{\alpha^{(\gamma)}}}{\partial x_j} + \frac{\partial q^{(\gamma)}}{\partial x_i} + \widetilde{\nu}_a^{(\gamma)} \frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial x_i} \\ &- \frac{\partial}{\partial x_j}\left(\widetilde{\nu}_a^{(\gamma)} \widetilde{\nu}^{(\gamma)} \frac{C_Y}{Y} \left(\frac{\partial v_i^{(\gamma)}}{\partial x_j} - \frac{\partial v_j^{(\gamma)}}{\partial x_i} \right) \right) + \rho^{(\gamma)} c_p^{(\gamma)} T_\alpha \frac{\partial T}{\partial x_i} \\ &+ \beta_{max} I\left(\xi^{(\gamma)}\right) u_i^{(\gamma)} = 0\,, \quad i = 1,2,3 \end{aligned} \tag{C1b}$$

$$\begin{aligned} R^{\widetilde{\nu}_a^{(\gamma)}} = &-\frac{\partial (v_j^{(\gamma)} \widetilde{\nu}_a^{(\gamma)})}{\partial x_j} - \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\widetilde{\nu}^{(\gamma)}}{\sigma} \right) \frac{\partial \widetilde{\nu}_a^{(\gamma)}}{\partial x_j} \right] + \frac{1}{\sigma} \frac{\partial \widetilde{\nu}_a^{(\gamma)}}{\partial x_j} \frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial x_j} \\ &+ 2\frac{c_{b2}}{\sigma} \frac{\partial}{\partial x_j}\left(\widetilde{\nu}_a^{(\gamma)} \frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial x_j} \right) + \widetilde{\nu}_a^{(\gamma)} \left(-P\left(\widetilde{\nu}^{(\gamma)}, \Delta\right) + D\left(\widetilde{\nu}^{(\gamma)}, \Delta\right) \right) \\ &+ \frac{\partial u_i^{(\gamma)}}{\partial x_j} \left(\frac{\partial v_i^{(\gamma)}}{\partial x_j} + \frac{\partial v_j^{(\gamma)}}{\partial x_i} \right) \frac{\partial v_t^{(\gamma)}}{\partial \widetilde{\nu}^{(\gamma)}} + \widetilde{\nu}^{(\gamma)} \widetilde{\nu}_a^{(\gamma)} C_{\widetilde{\nu}} \\ &+ \frac{\rho c_p}{Pr_t} \frac{\partial T_\alpha}{\partial x_j} \frac{\partial T}{\partial x_j} \frac{\partial v_t^{(\gamma)}}{\partial \widetilde{\nu}^{(\gamma)}} + \beta_{max} I\left(\xi^{(\gamma)}\right) \widetilde{\nu}_a^{(\gamma)} = 0 \end{aligned} \tag{C1c}$$

$$\begin{aligned} R^{\Delta_\alpha^{(\gamma)}} = &-2(1+\epsilon) \frac{\partial}{\partial x_j}\left(\frac{\partial \Delta}{\partial x_j} \Delta_\alpha^{(\gamma)} \right) - \epsilon \Delta \frac{\partial^2 \Delta_\alpha^{(\gamma)}}{\partial x_j^2} \\ &+ \widetilde{\nu}^{(\gamma)} \widetilde{\nu}_a^{(\gamma)} C_\Delta + \beta_{max} I\left(\xi^{(1)} + \xi^{(2)}\right) \Delta_\alpha^{(\gamma)} = 0 \end{aligned} \tag{C1d}$$

$$R^{T_\alpha} = -\sum_{\gamma=1,2} \rho^{(\gamma)} c_p^{(\gamma)} \frac{\partial (v_j^{(\gamma)} T_\alpha)}{\partial x_j} - \frac{\partial}{\partial x_j}\left(k_{eff} \frac{\partial T_\alpha}{\partial x_j} \right) = 0 \tag{C1e}$$

Here, $\tau_{ij}^{\alpha^{(\gamma)}} = \left(\nu^{(\gamma)} + \nu_t^{(\gamma)}\right)\left(\frac{\partial u_i^{(\gamma)}}{\partial x_j} + \frac{\partial u_j^{(\gamma)}}{\partial x_i} \right)$ are the adjoint stress tensors. Terms C_Y, $C_{\widetilde{\nu}}$ and C_Δ emerge from the differentiation of the production and dissipation terms of the turbulence model and read, Papoutsis-Kiachagias et al. (2019),

$$C_Y = -\left(c_{b1} + c_{w1} \mathcal{C} \frac{r}{\widetilde{Y}} \right) \tag{C2a}$$

$$C_{\widetilde{\nu}} = -\left(c_{b1} + c_{w1} \mathcal{C} \frac{\widetilde{\nu}}{\Delta^2} \right)\left(\frac{f_{v2}}{\kappa^2 \Delta^2} + \frac{\widetilde{\nu}}{\kappa^2 \Delta^2} \frac{\partial f_{v2}}{\partial \widetilde{\nu}} \right) + c_{w1} \mathcal{C} \frac{r}{\Delta^2} + c_{w1} \frac{f_w}{\Delta^2} \tag{C2b}$$

$$C_\Delta = -\frac{2}{\Delta^3}\left[c_{w1} r \mathcal{C} \frac{\widetilde{\nu}}{\Delta^2}\left(\Delta^2 - \frac{\widetilde{\nu} f_{v2}}{\kappa^2 \widetilde{Y}} \right) + c_{w1} f_w \widetilde{\nu} - c_{b1} \frac{f_{v2}}{\kappa^2} \widetilde{\nu} \right] \tag{C2c}$$

where

$$\mathcal{C} = \left[1 + c_{w2}\left(6r^5 - 1\right)\right] \frac{c_{w3}^6}{g^6 + c_{w3}^6} \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{\frac{1}{6}}$$

$$\frac{\partial f_{v1}}{\partial \widetilde{\nu}} = \frac{1}{\nu} \frac{3\chi^2 c_{v1}^3}{\left(\chi^3 + c_{v1}^3\right)}$$

$$\frac{\partial f_{v2}}{\partial \widetilde{\nu}} = \frac{1 - \chi \frac{\partial f_{v1}}{\partial \widetilde{\nu}}}{\left(1 + \chi f_{v1}\right)^2}$$

Regarding numerical implementation, the adjoint PDEs are discretized using the finite volume method and solved using a pressure-correction approach, Patankar (1980), similarly to their primal counterpart. Since the adjoint energy equation does not depend on any other adjoint variable apart from T_α, it is solved prior to the adjoint mean-flow equations for each fluid (Eqs. C1a–C1c). Finally, the adjoint distance $\Delta_\alpha^{(\gamma)}$ does not appear in Eqs. C1a, C1b, C1c and C1e. Therefore, Eq. C1d can be solved last. Second-order accurate upwind and central schemes are used for the discretization of convection and diffusion terms, respectively.

The *BC* terms, which appear in some surface integrals of Eq. 20, are associated with the boundary conditions of the adjoint variables and read

$$BC_1 = u_i^{(\gamma)} n_i + \frac{\partial J}{\partial p^{(\gamma)}} \tag{C3a}$$

$$\begin{aligned} BC_{2,i} = &\, v_j^{(\gamma)} n_j u_i^{(\gamma)} + \tau_{ij}^{\alpha^{(\gamma)}} n_j - q^{(\gamma)} n_i + \widetilde{\nu}_a^{(\gamma)} \widetilde{\nu}^{(\gamma)} \frac{C_Y}{Y}\left(\frac{\partial v_i^{(\gamma)}}{\partial x_j} - \frac{\partial v_j^{(\gamma)}}{\partial x_i} \right) n_j \\ &+ \frac{\partial J}{\partial v_i^{(\gamma)}}\,, \quad i = 1,2,3 \end{aligned} \tag{C3b}$$

$$BC_3 = v_j^{(\gamma)} n_j \widetilde{\nu}_a^{(\gamma)} + \left(\nu + \frac{\widetilde{\nu}^{(\gamma)}}{\sigma} \right) \frac{\partial \widetilde{\nu}_a^{(\gamma)}}{\partial n} + \frac{\partial J}{\partial \widetilde{\nu}^{(\gamma)}} \tag{C3c}$$

$$BC_4 = (2+\epsilon) \Delta_\alpha^{(\gamma)} \frac{\partial \Delta}{\partial n} + \epsilon \Delta \frac{\partial \Delta_\alpha^{(\gamma)}}{\partial n} \tag{C3d}$$

$$BC_5 = \sum_{\gamma=1,2} \rho^{(\gamma)} c_p^{(\gamma)} v_j^{(\gamma)} n_j T_\alpha + k_{eff} \frac{\partial T_\alpha}{\partial n} + \frac{\partial J}{\partial T} \tag{C3e}$$

The adjoint boundary conditions depend on those of the primal problem and can be found in Table 3. Their derivation is described in detail in Papoutsis-Kiachagias and Giannakoglou (2016); Kontoleontos et al. (2013); Gkaragkounis et al. (2018).

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Table 3 Primal and adjoint boundary conditions at the inlets, outlets and external wall boundaries of the heat exchanger

	Inlets $S_I^{(\gamma)}$	External Walls $S_W^{(\gamma)}$	Outlets $S_O^{(\gamma)}$
$v_i^{(\gamma)}$	$v_i^{(\gamma)} = v_{0,i}^{(\gamma)}$	$v_i^{(\gamma)} = 0$	$\frac{\partial v_i^{(\gamma)}}{\partial n} = 0$
$p^{(\gamma)}$	$\frac{\partial p^{(\gamma)}}{\partial n} = 0$	$\frac{\partial p^{(\gamma)}}{\partial n} = 0$	$p^{(\gamma)} = 0$
$\widetilde{\nu}^{(\gamma)}$	$\widetilde{\nu}^{(\gamma)} = \widetilde{\nu}_0$	$\widetilde{\nu}^{(\gamma)} = 0$	$\frac{\partial \widetilde{\nu}^{(\gamma)}}{\partial n} = 0$
T	$T = T_I$	$k_{eff}\frac{\partial T}{\partial n} = 0$	$\frac{\partial T}{\partial n} = 0$
Δ	$\frac{\partial \Delta}{\partial n} = 0$	$\Delta = 0$	$\frac{\partial \Delta}{\partial n} = 0$
$u_i^{(\gamma)}$	$BC_1 = 0\,,\ u_i^{(\gamma)} t_i = 0$	$BC_1 = 0\,,\ u_i^{(\gamma)} t_i = 0$	$\frac{\partial u_{\langle n \rangle}^{(\gamma)}}{\partial n} = 0\,,\ BC_{2,i} t_i = 0$
$q^{(\gamma)}$	$\frac{\partial q^{(\gamma)}}{\partial n} = 0$	$\frac{\partial q^{(\gamma)}}{\partial n} = 0$	$BC_{2,i} n_i = 0$
$\widetilde{\nu}_a^{(\gamma)}$	$\widetilde{\nu}_a^{(\gamma)} = 0$	$\widetilde{\nu}_a^{(\gamma)} = 0$	$BC_3 = 0$
T_α	$T_\alpha = 0$	$\frac{\partial T_\alpha}{\partial n} = 0$	$BC_5 = 0$
$\Delta_\alpha^{(\gamma)}$	$\frac{\partial \Delta_\alpha^{(\gamma)}}{\partial n} = 0$	$\Delta_\alpha^{(\gamma)} = 0$	$\frac{\partial \Delta_\alpha^{(\gamma)}}{\partial n} = 0$

Acknowledgements The first three authors are indebted to Mitsubishi Heavy Industries LTD. (MHI), Japan, for funding part of this research. They, also, feel the need to acknowledge Dr. Konstantinos Gkaragkounis, NTUA, for his contributions in the development of the adjoint-based CHT ShpO software.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Replication of results All geometries and flow conditions for the studied cases are described analytically.

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